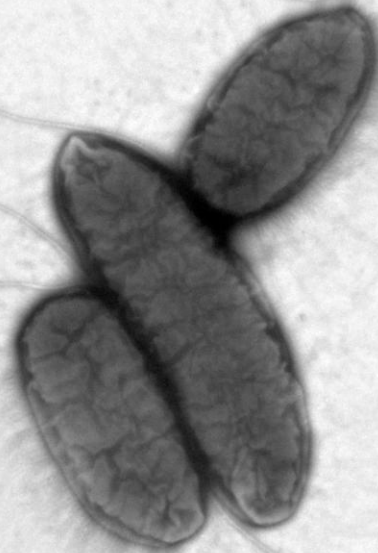


Bakterie struktur



Svend Birkelund 2026

Studieordning 2024

Disposition

- Grams farvemetode
- Peptidoglycanlag syntese
- Lipopolysaccharid
- Form og lejring af bakterier
- Organeller

Taksonomi

- Domænet "Eukaryote" har cellekerne
- Domænerne "Bakterier" og "Archaea" har ikke cellekerne m.m.
- I domænet "Archaea" er der f.eks. termotabile bakterier som *Thermus aquaticus*, der lever ved temperaturer over 70 °C
 - Hvorfra *Taq* polymerase er fremstillet
- I domænet "Bacteria" findes bl.a. de human patogene bakterier

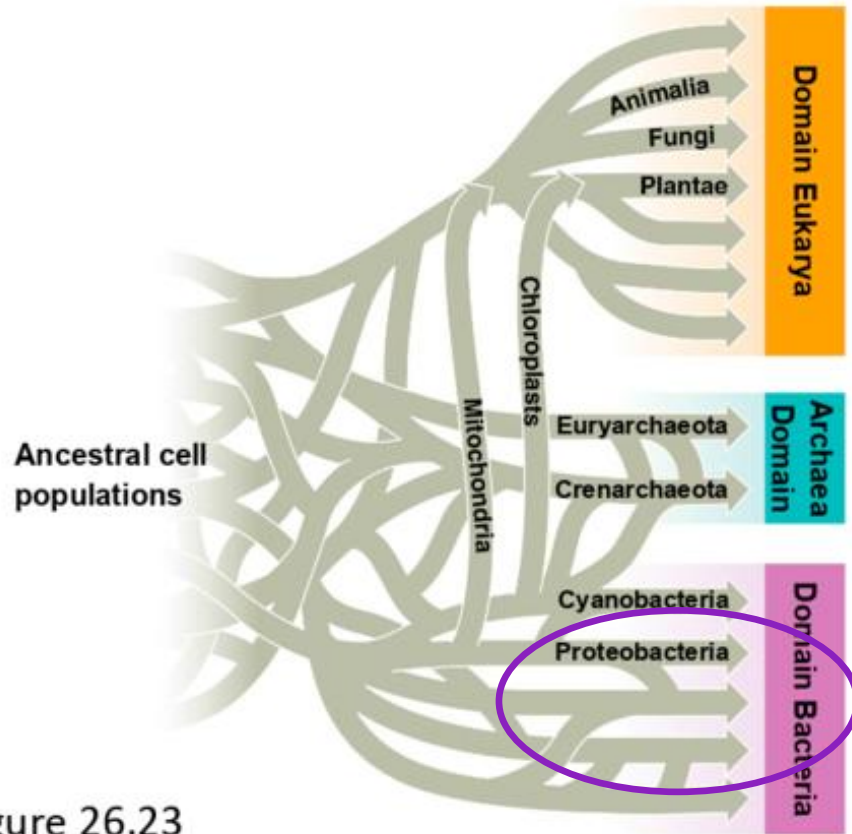
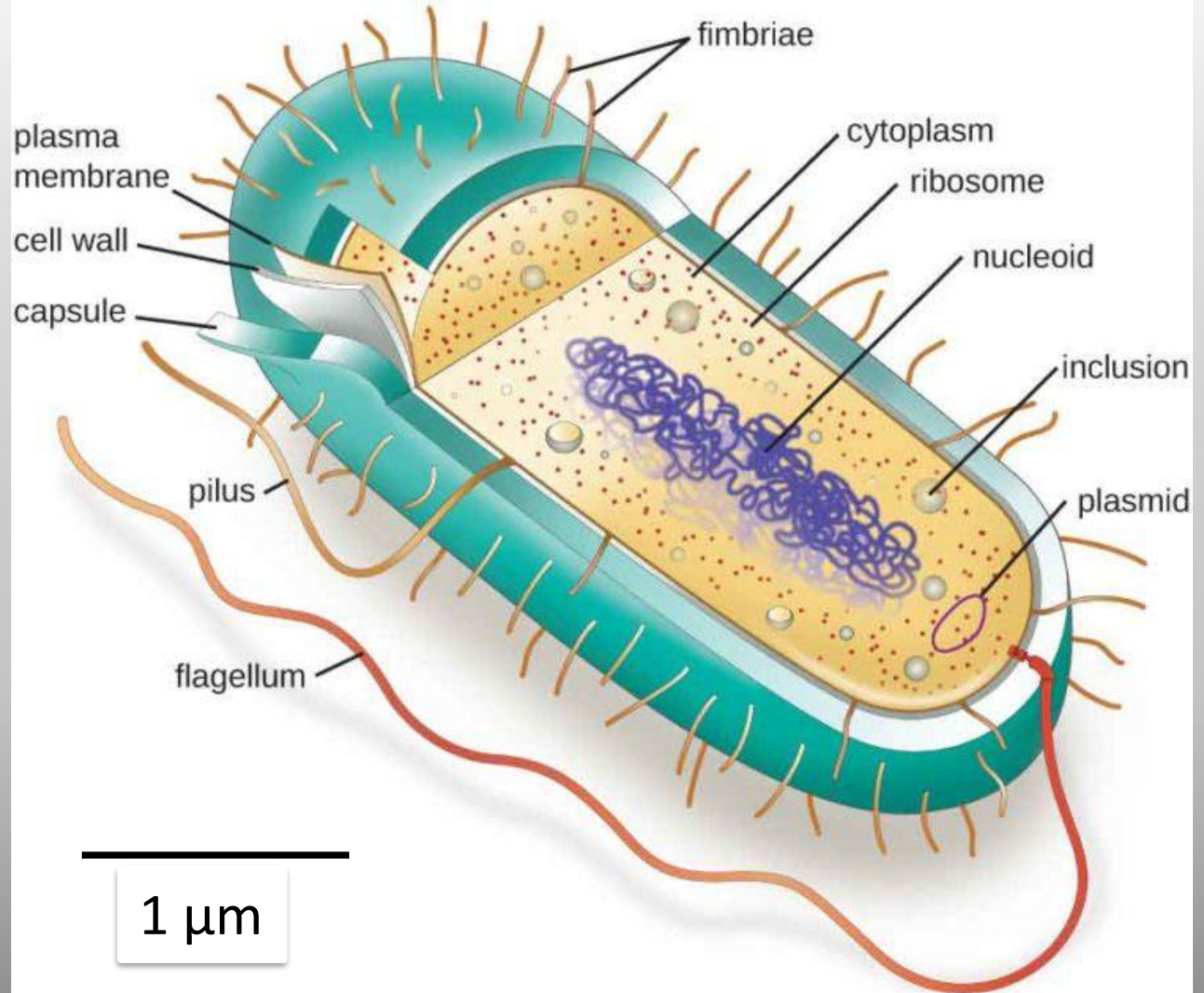


Figure 26.23

Prokaryot mikroorganisme



Eukaryot celle *Candida albicans* (svamp) Staphylococcus aureus (bakterie)

Nomenklatur er latinsk med *kursiv*
lige som i anatomi

Musculus quadriceps

M. quadriceps

Bakterier

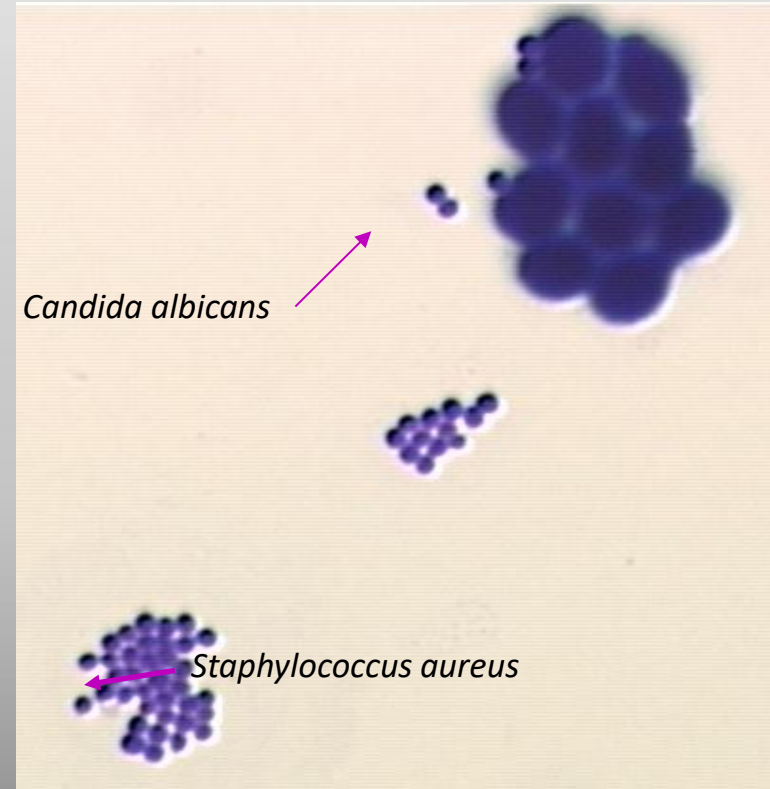
Genus species

Staphylococcus aureus

S. aureus

Familie *Genus species*

Enterobacteriaceae *Escherichia coli*



Hvorfor er *Genus species* væsentligt

- *Streptococcus pneumoniae* ➤ *S. pneumoniae*
- *Mycoplasma pneumoniae* ➤ *M. pneumoniae*
- *Chlamydia pneumoniae* ➤ *C. pneumoniae*
- *Klebsiella pneumoniae* ➤ *K. pneumoniae*
 - Obs: giver kun sjældende pneumonier

Species navn er tit opkaldt efter sygdommen bakterien giver,

- *Neisseria meningitidis*
- *Mycobacterium tuberculosis*

men bakterien kan også tit give andre sygdomme

Bakterien *Haemophilus influenzae* blev isoleret fra mange patienter under den spanske syge i 1918, men den giver **ikke** influenza, men opportunistiske- eller superinfektioner.

Den behold dog sit navn selvom man senere fandt ud af at det var en virus der gav influenza

Forskelle mellem pro- og eukaryote

➤ **Prokaryote**

- DNA frit
- DNA -> RNA -> Protein
 - ingen introns
- 70S (16S/23S) ribosomer
- Ingen organeller i cytoplasma
- Cellevæg
- Gram positiv/negativ.
- Kapsel

➤ **Eukaryote human celle**

- DNA i kerne
- DNA -> RNA processing -> translation
 - introns
- 80S (40S/60S) ribosomer
- Organeller med membraner, lysosomer, mitokondrier m.m.
- Ingen Cellevæg (humane celler)

Dyrkningsmedier

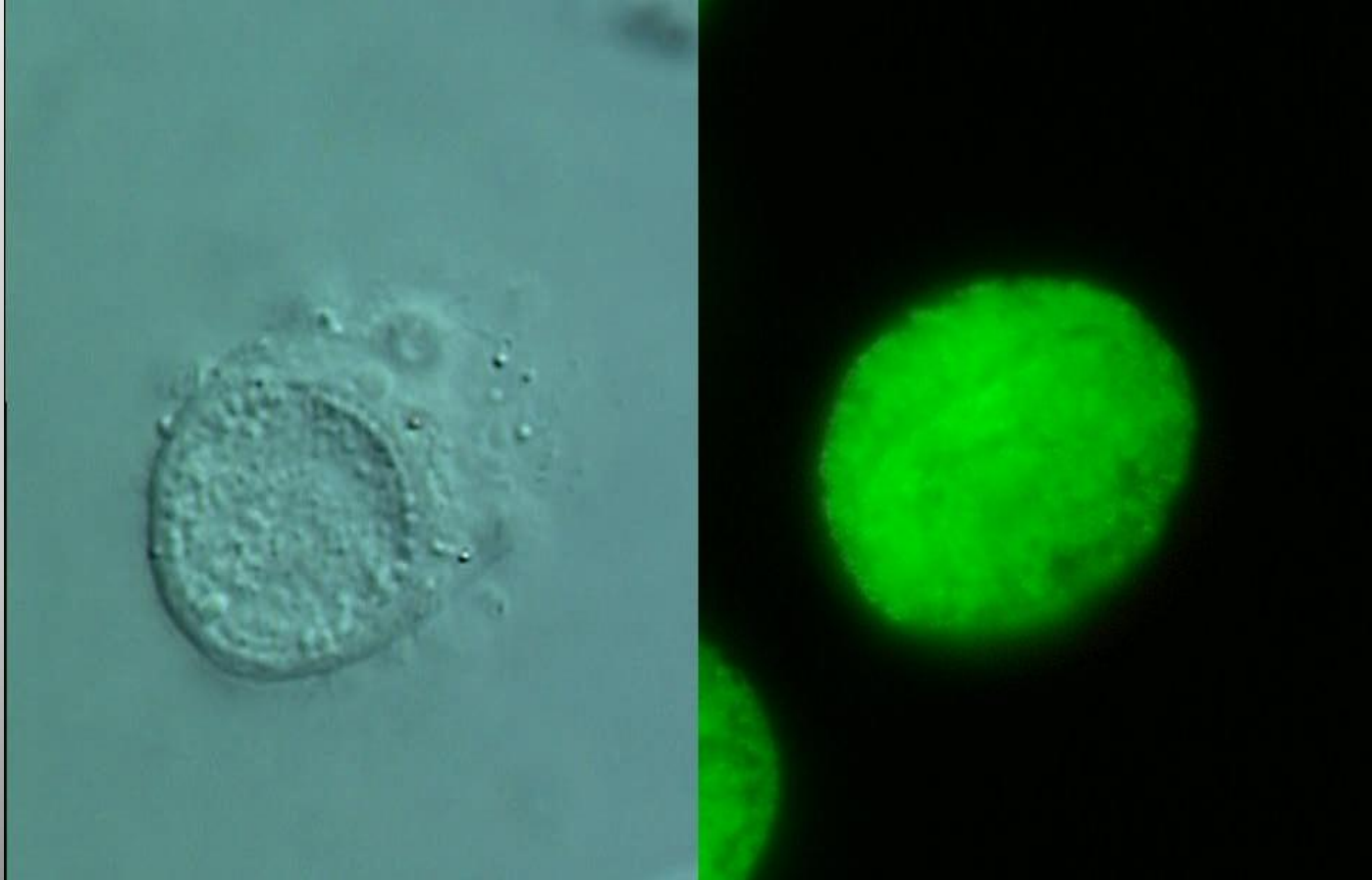
- **Blodplader (Hæmolyse) ikke selektive**
- **Chokolade agar plader (hæmolyseret blod), beriget medium**
- **MacConkey agar, selektiv, indikator**
- **Müller Hinton**
- **Special medier**
 - Løvenstein-Jensens substrat (malakitgrønt), selektivt medium for *M. tuberculosis*
 - Cellekultur (til obligat intracellulære bakterier)

Blodplade
Staphylococcus
aureus
 β -haemolyse





***Chlamydia trachomatis* serovar D**



Grams farvemetode

Grams farvning

- Hans Christian Joachim Gram (13 September, 1853 – 14 November, 1938)
- Botanik studerende – videnskabelig assistent – Isolering af farmaka fra planter
- Medicinstudium 1878-83
- Europarejser 1878-85
- I 1884 udviklede han en farvemetode i Berlin til at adskille bakterier – "Grams farvemetode"
- Karakteriserede også sygdommen pernicios anæmi
- Professor, senere Medicinsk overlæge København
- Pensioneret 1923
- Transmissions-elektronmikroskopet blev udviklet i 1930'erne, dermed kunne forskellen mellem gram negative og positive bakterier vises strukturelt

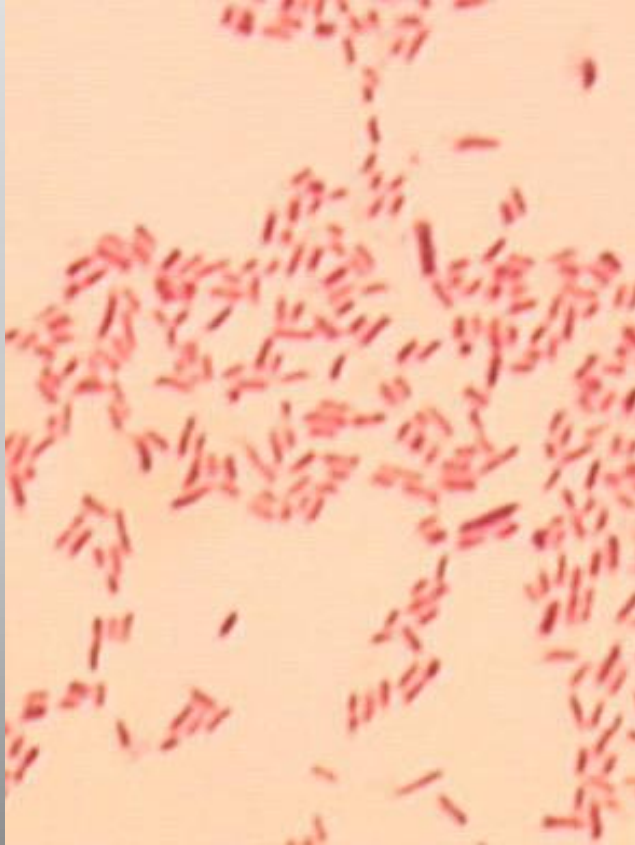


Gram farvning (Video)

	Positiv	Negativ
Lufttørring, Varmefiksering		
Crystal violet		
Jodjodkalium		
Vask med Alkohol > Vand		
Safranin rød		
Affarvning med vand		

Gram positiv/negativ

Escherichia coli



Streptococcus pneumoniae

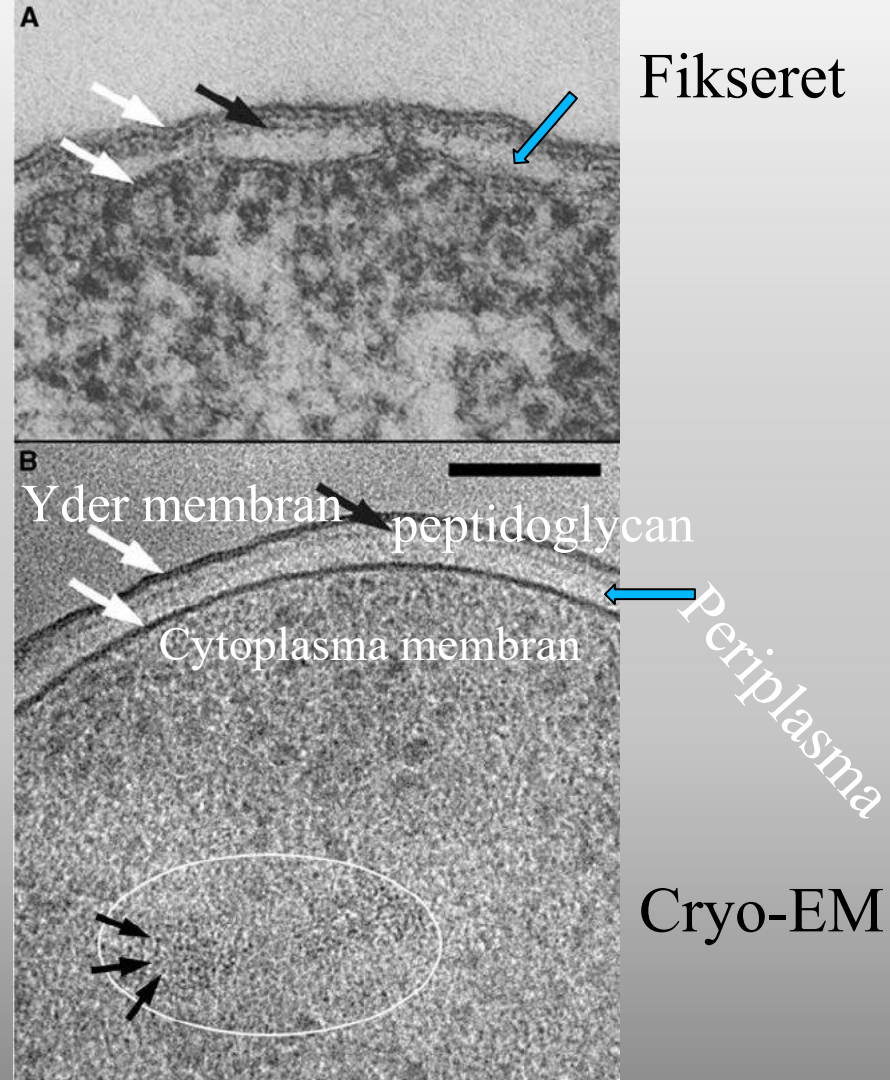


Gram negative bacteria

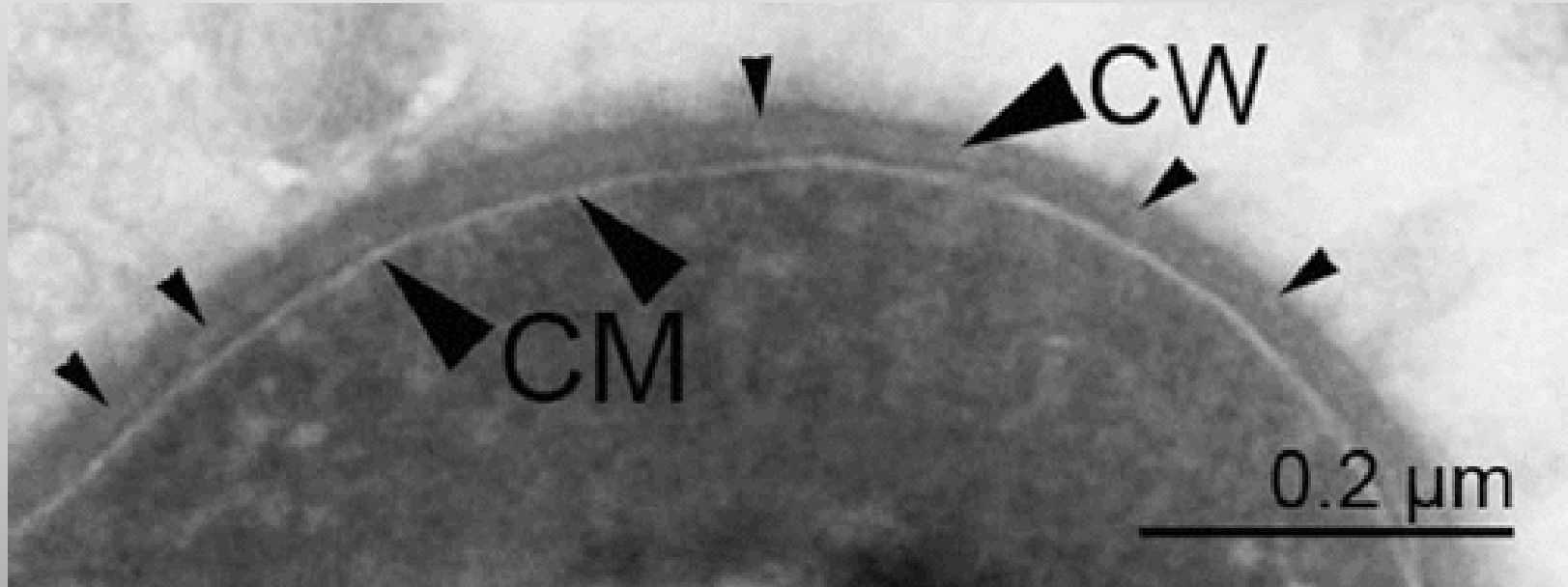
Escherichia coli

- **Cryo-electron microscopy of vitreous sections**

- The EMBO Journal 2004, 23:3583-8
- Opbygning
 - Ydermembran
 - Peptidoglycanlag
 - Periplasma
 - Cytoplasma membran



Gram positiv Cryo-EM Streptococcus pneumoniae



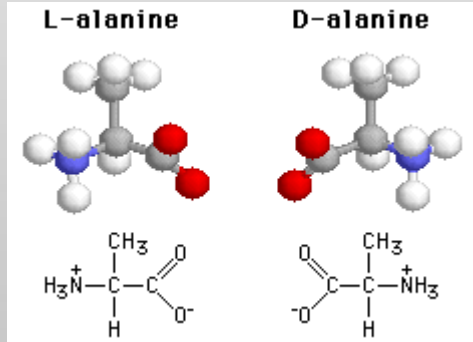
CM: cytoplasmic membrane CW: cell wall – peptidoglycan layer

DOI: <https://doi.org/10.1128/microbiolspec.gpp3-0044-2>

Peptidoglycanlag syntese

Proteiner indeholder **kun** L-aminosyre i både eukaryote og prokaryote

Bakteria har Alanin-racemase



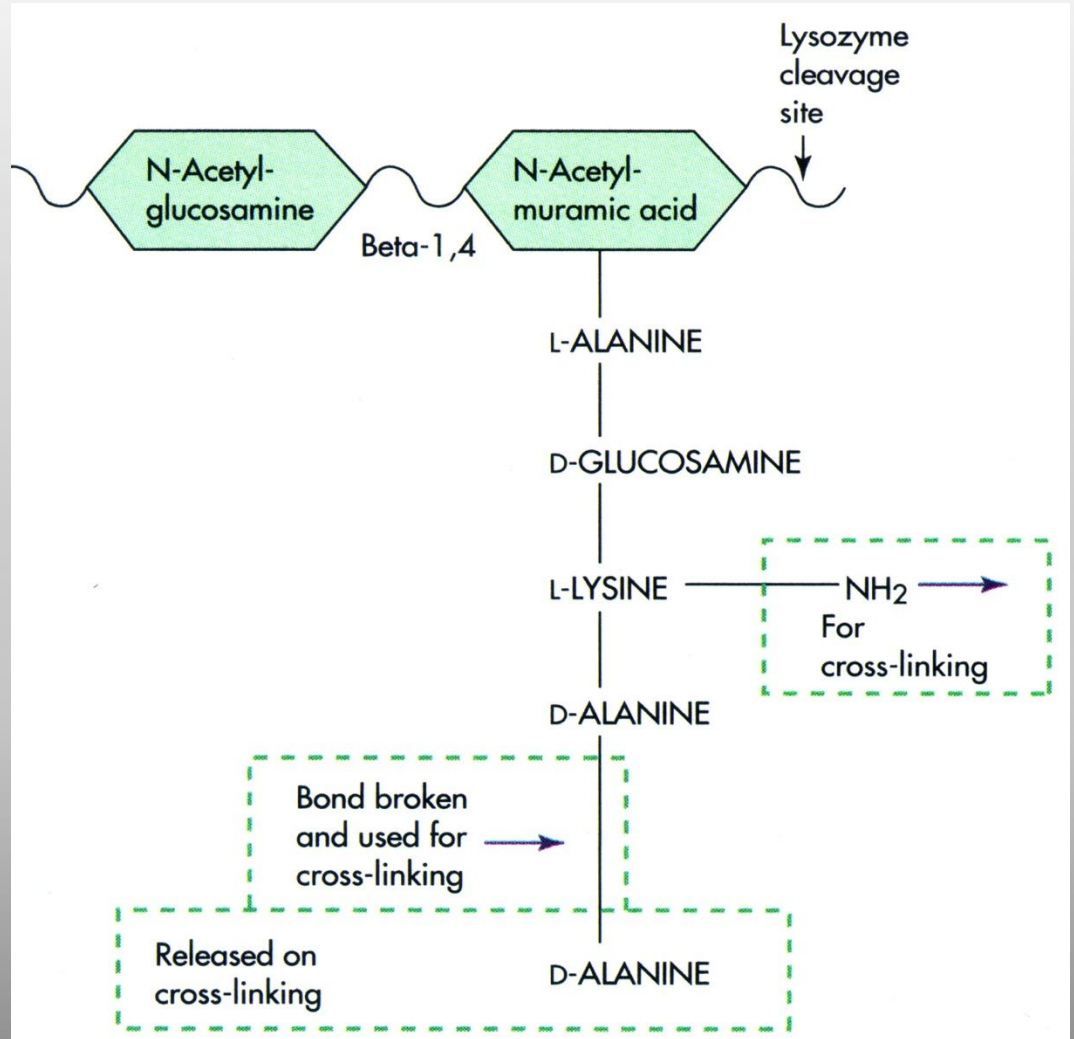
stereoisomer aminosyre

Laevus (lat.); venstre (da.)

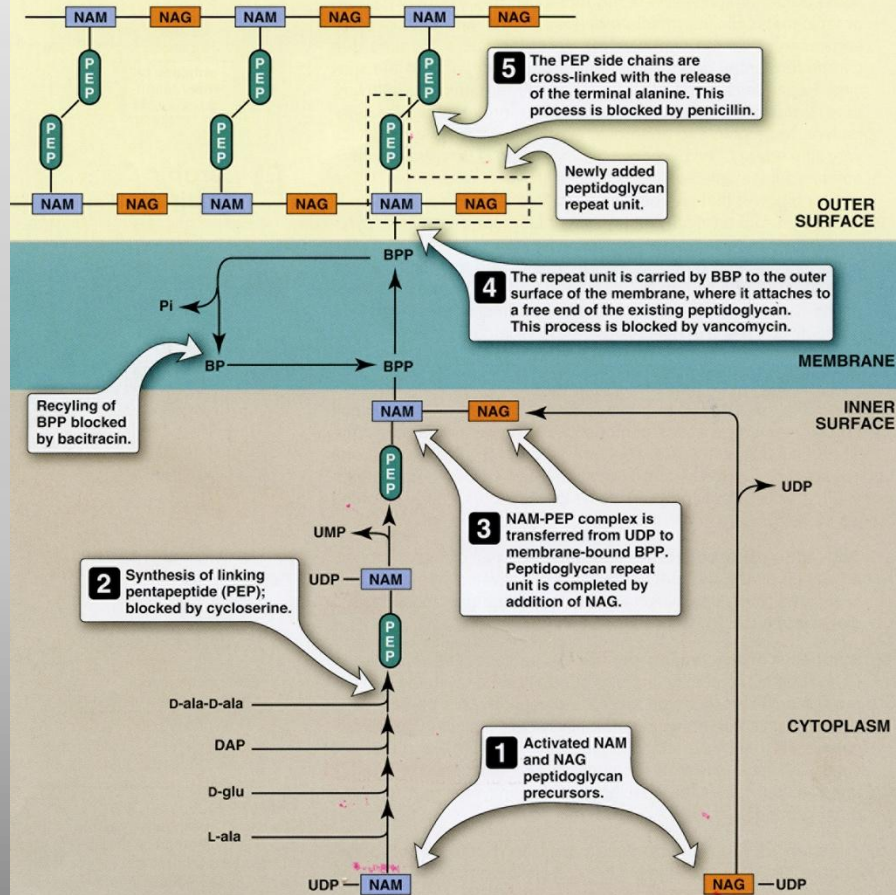
Dexter (lat.); højre (da.)

Sinister (lat.);

venstre (medicinsk latin)



NAM	=	N-acetylmuramic acid
NAG	=	N-acetylglucosamine
BPP	=	Bactoprenol phosphate
PEP	=	Cross-linking peptide
DAP	=	Diaminopimelic acid

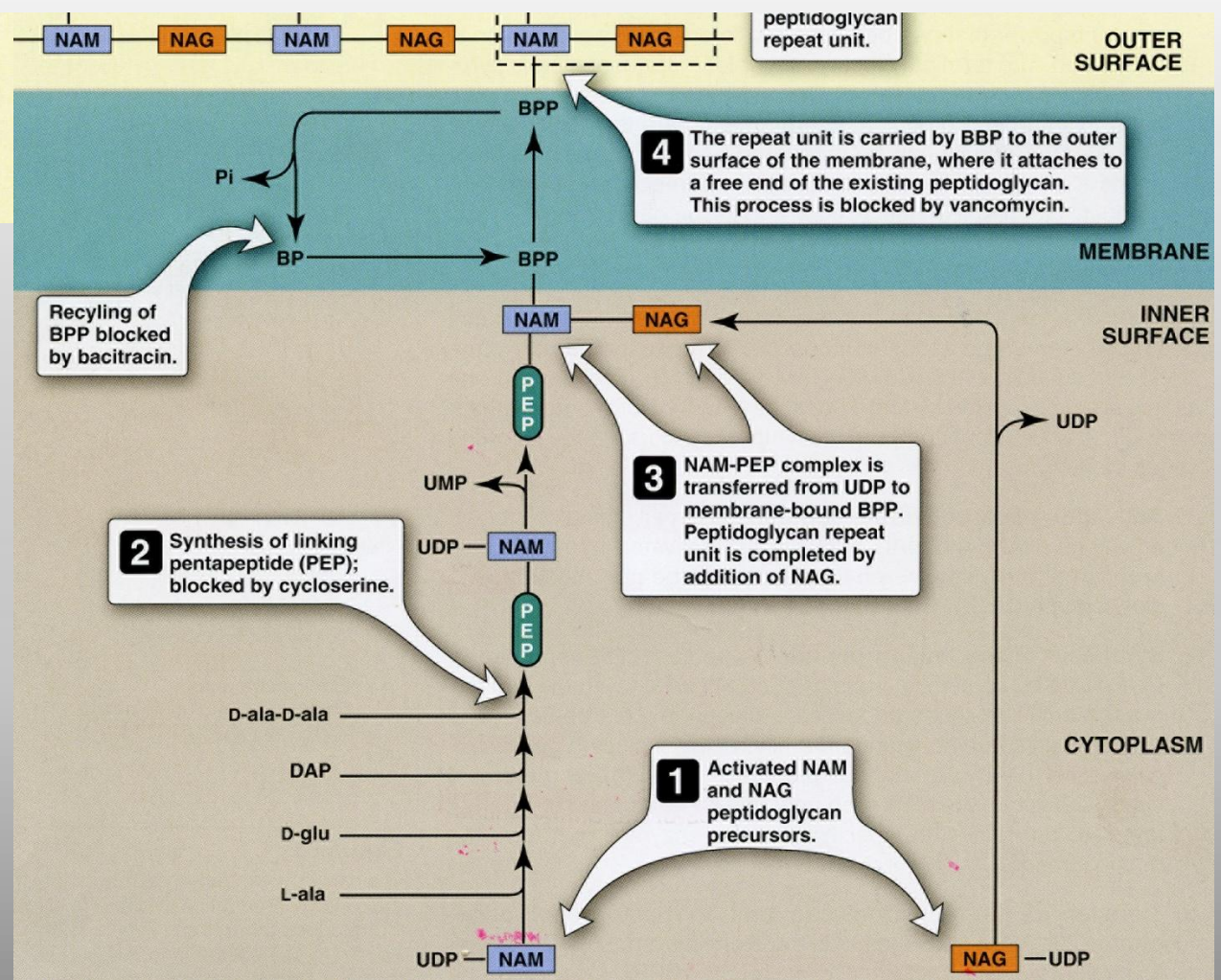


NAM	=	<i>N</i> -acetylmuramic acid
NAG	=	<i>N</i> -acetylglucosamine
BPP	=	Bactoprenol phosphate
PEP	=	Cross-linking peptide
DAP	=	Diaminopimelic acid

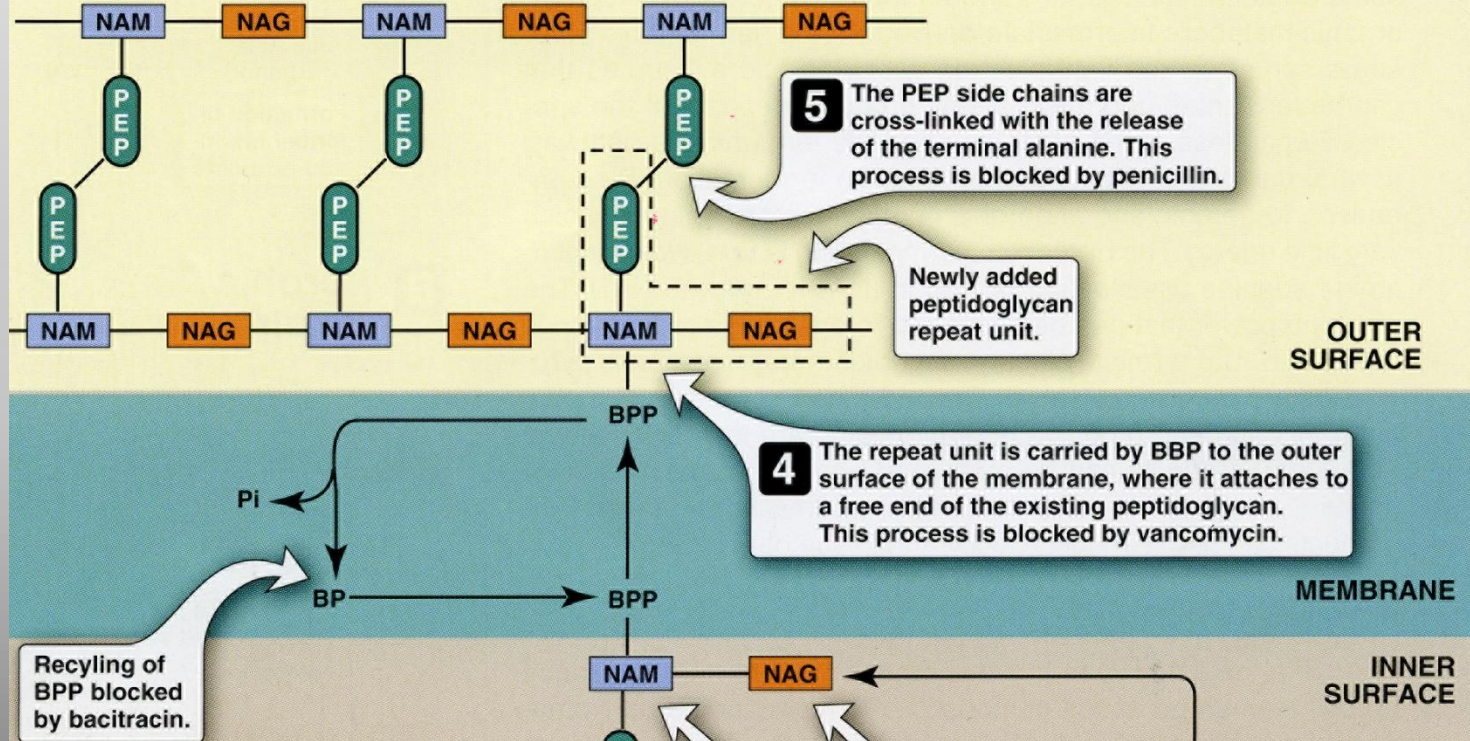
UTP

Uridine

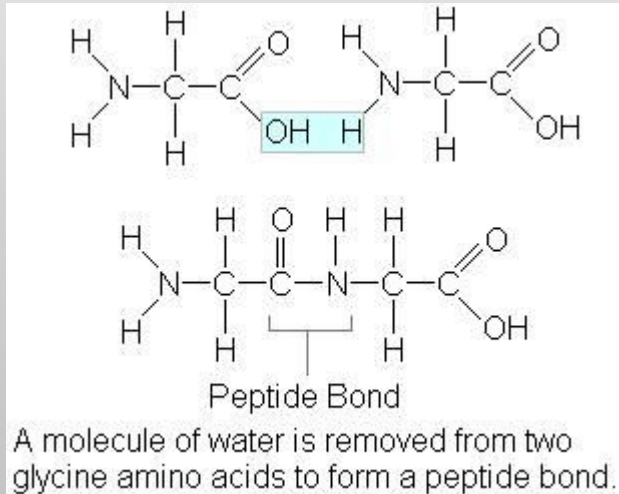
triphosphate



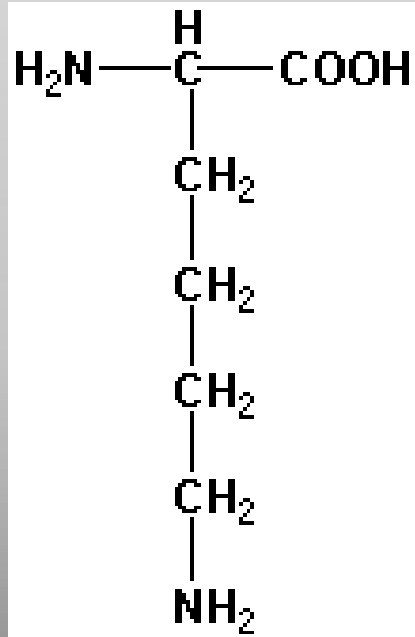
NAM	=	<i>N</i> -acetylmuramic acid
NAG	=	<i>N</i> -acetylglucosamine
BPP	=	Bactoprenol phosphate
PEP	=	Cross-linking peptide
DAP	=	Diaminopimelic acid



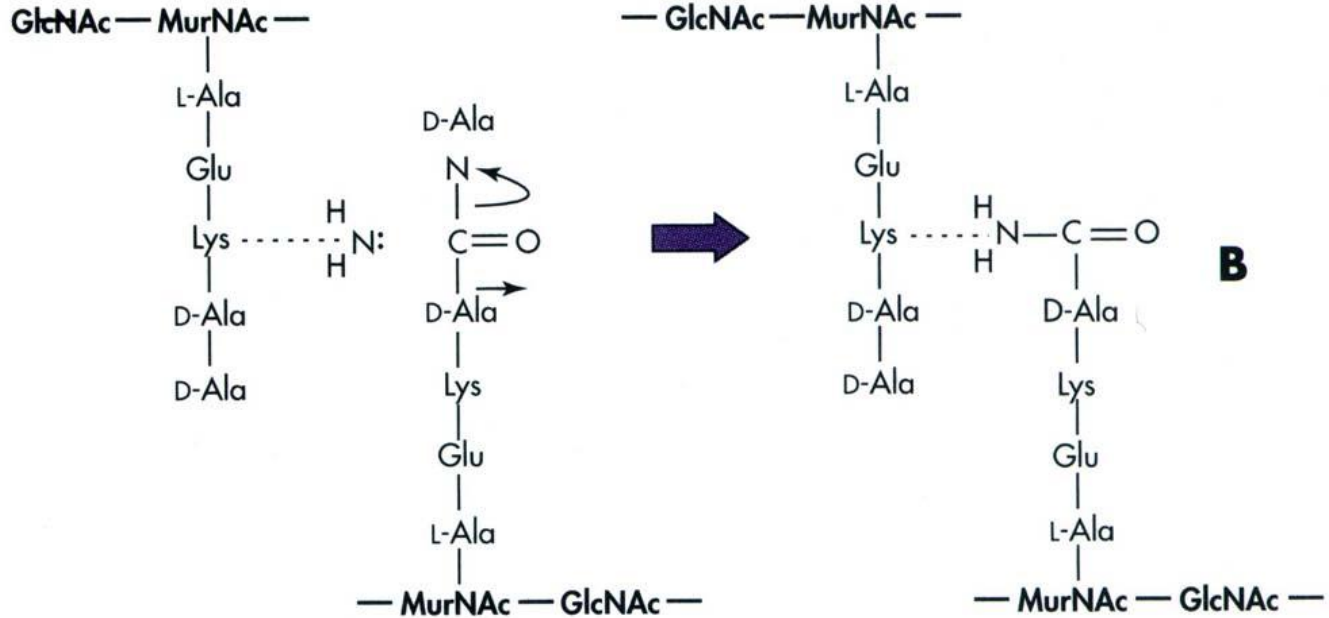
Peptid binding



Lysin



Transpeptidation Reaction

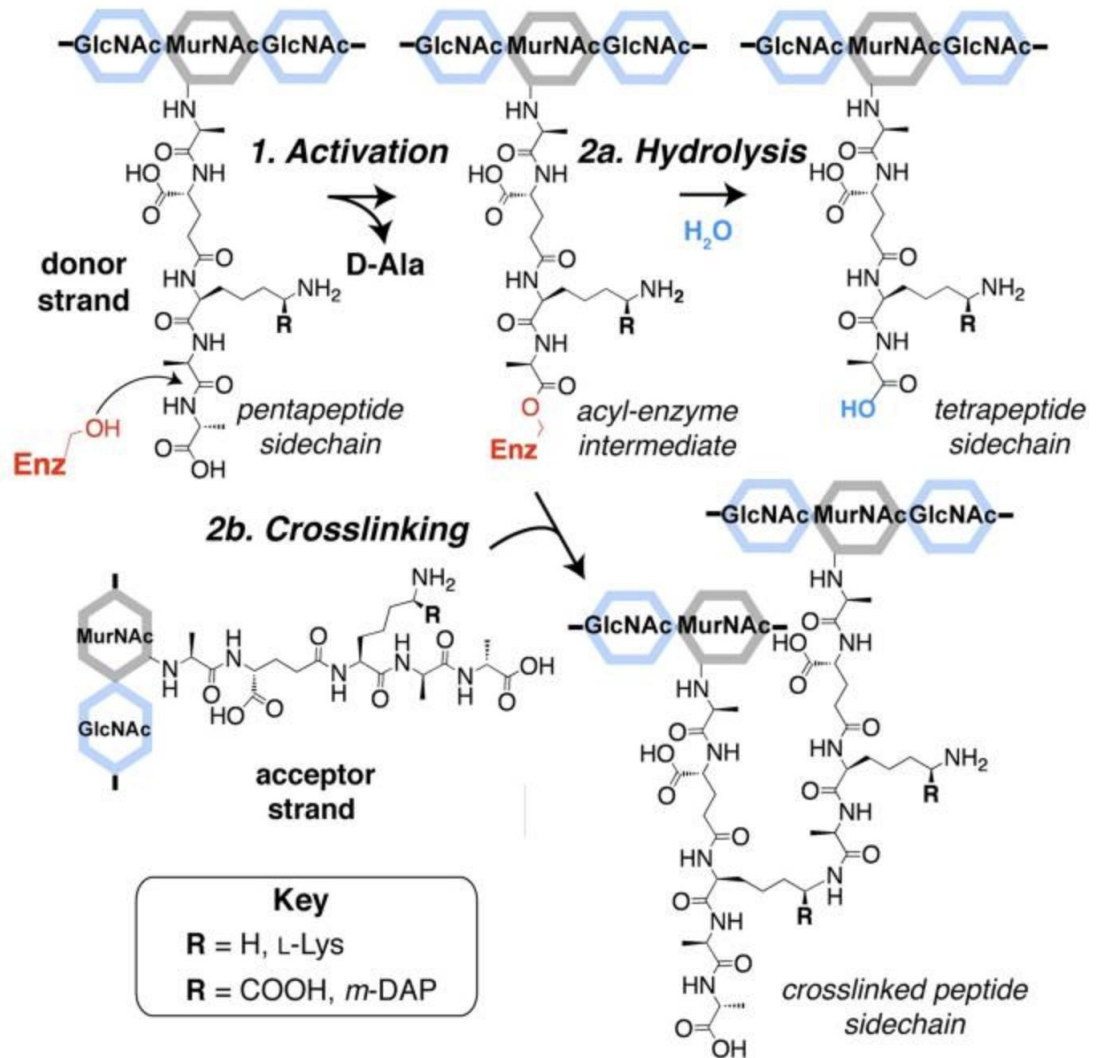


Transpeptidase har en unik enzymatisk reaktion med et kovalent intermediat stadium

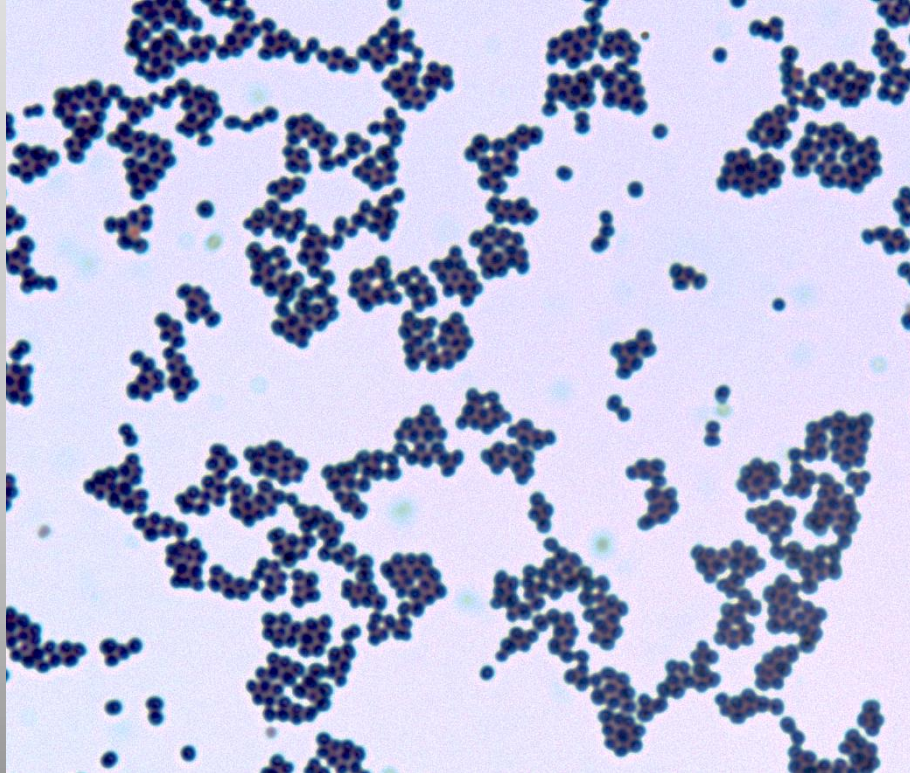
1) PBP binder sig kovalent og acyl-PBP binding dannes under fraspaltning af terminal D-alanin

2b) acyl-PBP reagerer med NH_2 i lysin sidekæde, PBP frigives

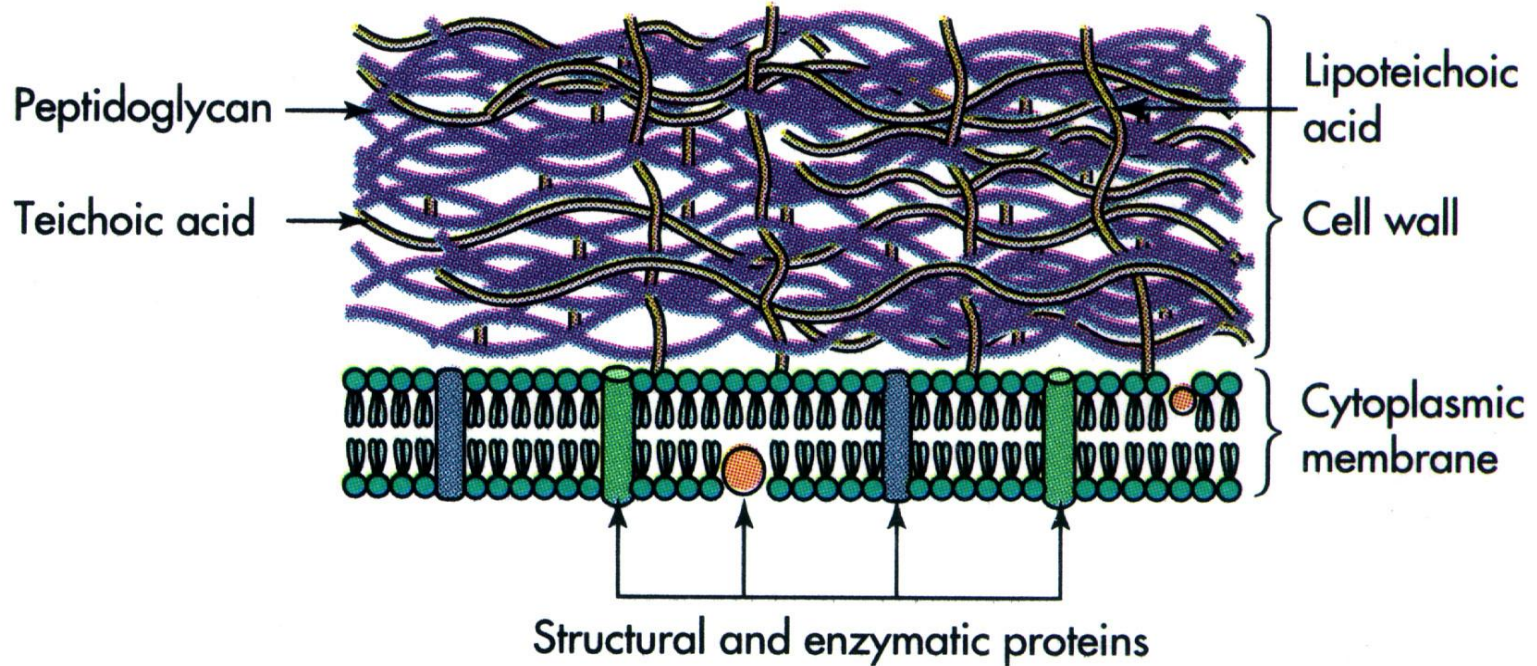
2a) PBP frigives ved hydrolyse



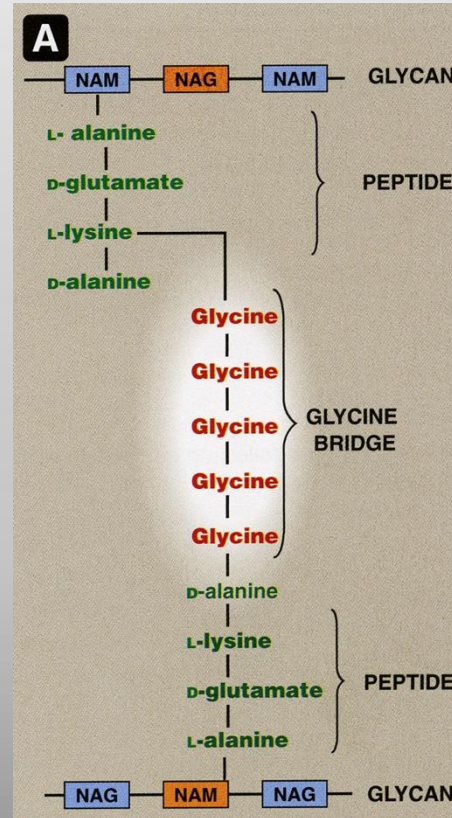
Gram farvning *Staphylococcus aureus*



Gram positiv cellevæg



Specielt for gram positive bakterier



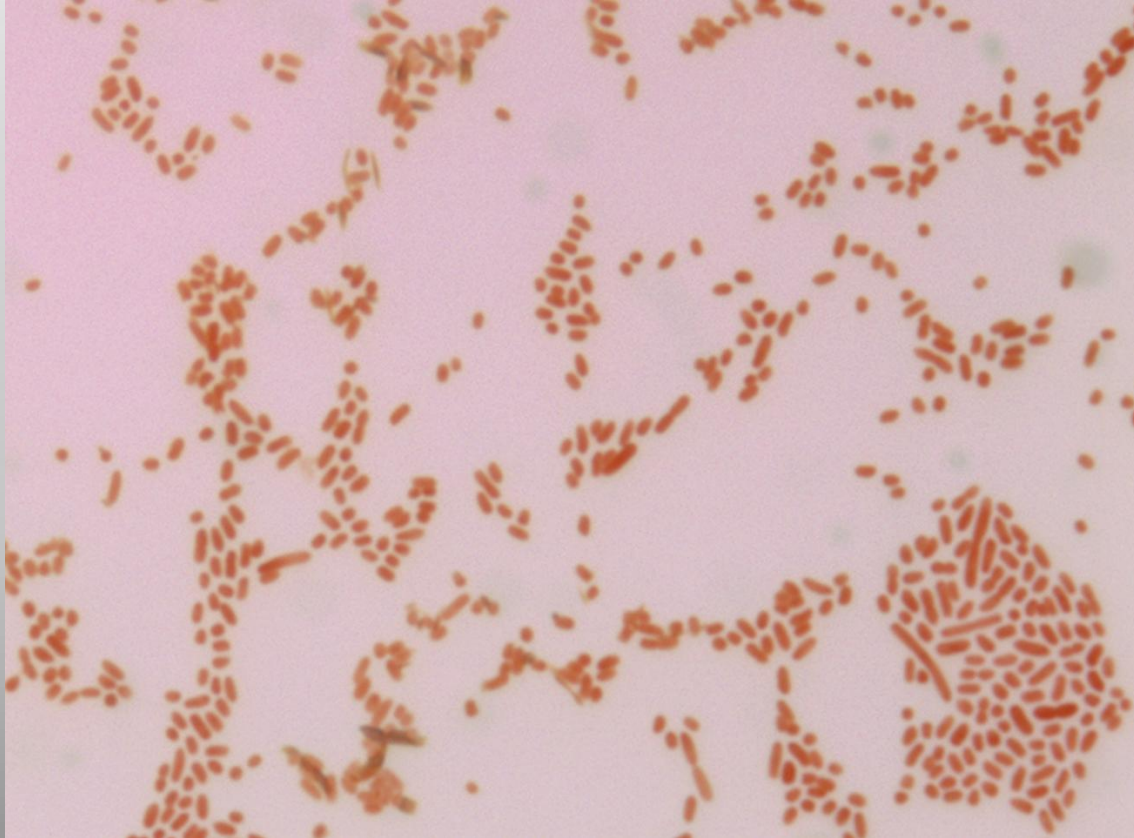
Peptidoglycan lag

- N-acetylglucosamin (NAG) og N-acetylmuramic acid (NAM) kæde
- NAM har side kæde med pentavalent D- og L-aminosyre
- Pentavalent sidekæder krydsbindes af Penicillin-bindende proteiner under fraspaltning af D-Ala,
- PBP har en kovalent intermitter i den enzymatisk reaktion
- Gram negative bakterier, tyndt peptidoglycan lag i periplastisk rum
- Gram positive bakterier, tykt peptidoglycan lag med 5' mer glycin linker

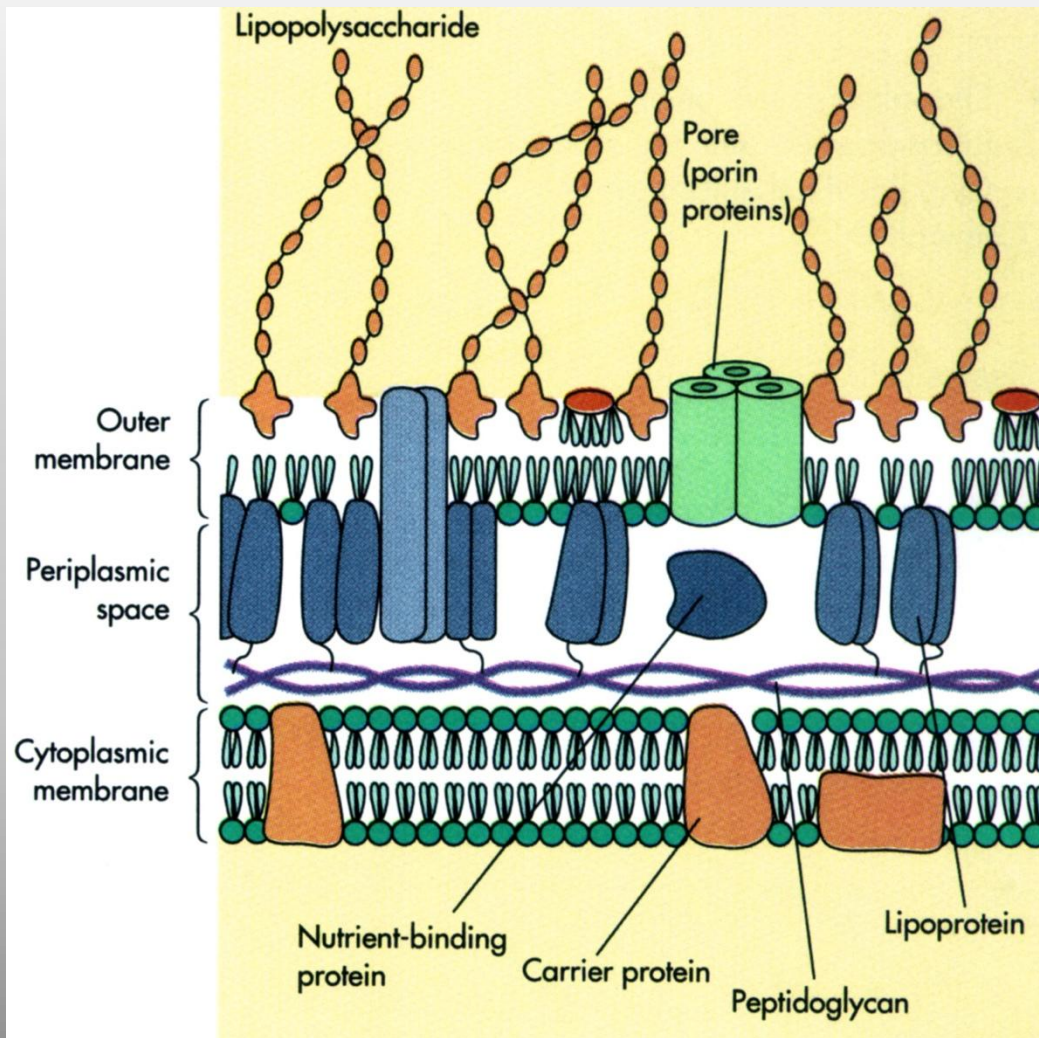
A



Gram farvning *Proteus mirabilis*



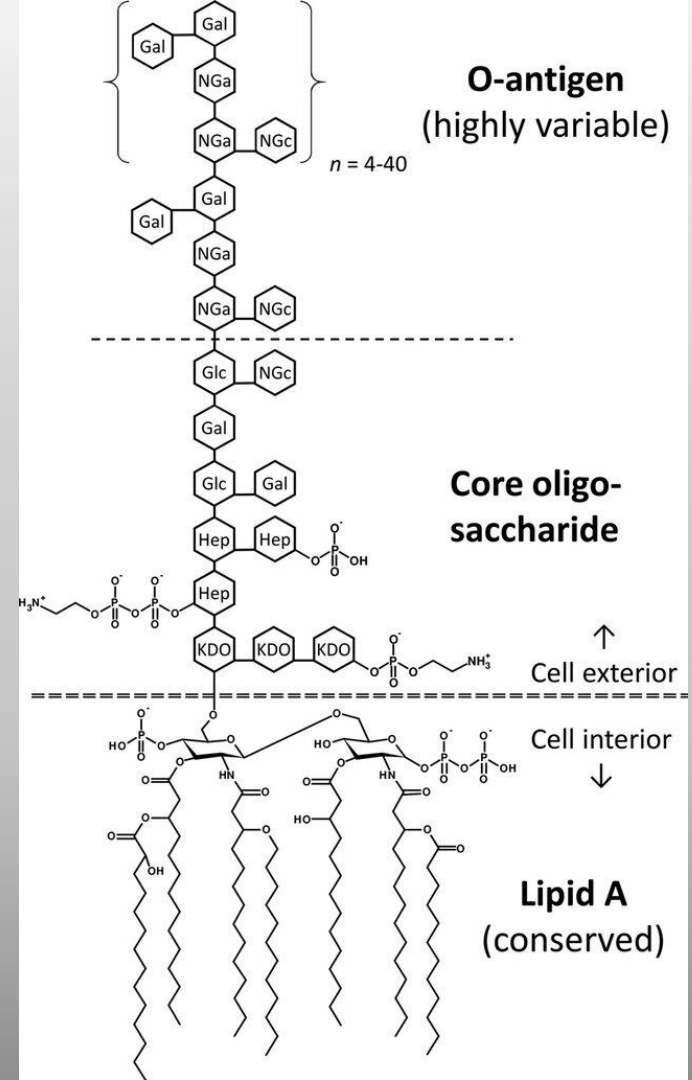
Gram negativ cellevæg



Lipopolysaccharid (LPS)

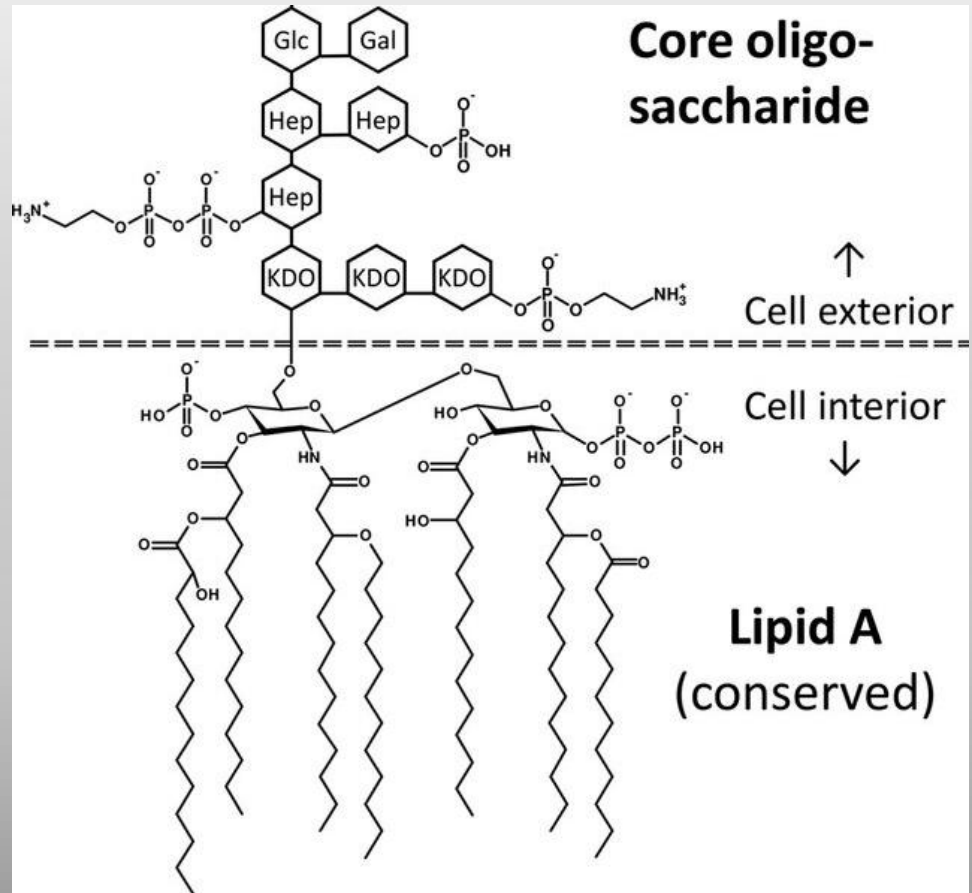
Ældre navn "endotoksin" pga. den fysiologisk virkning. Dette navn bruges stadig ofte.

Adams et al., Biophys. J. 106, 2395 (2014)

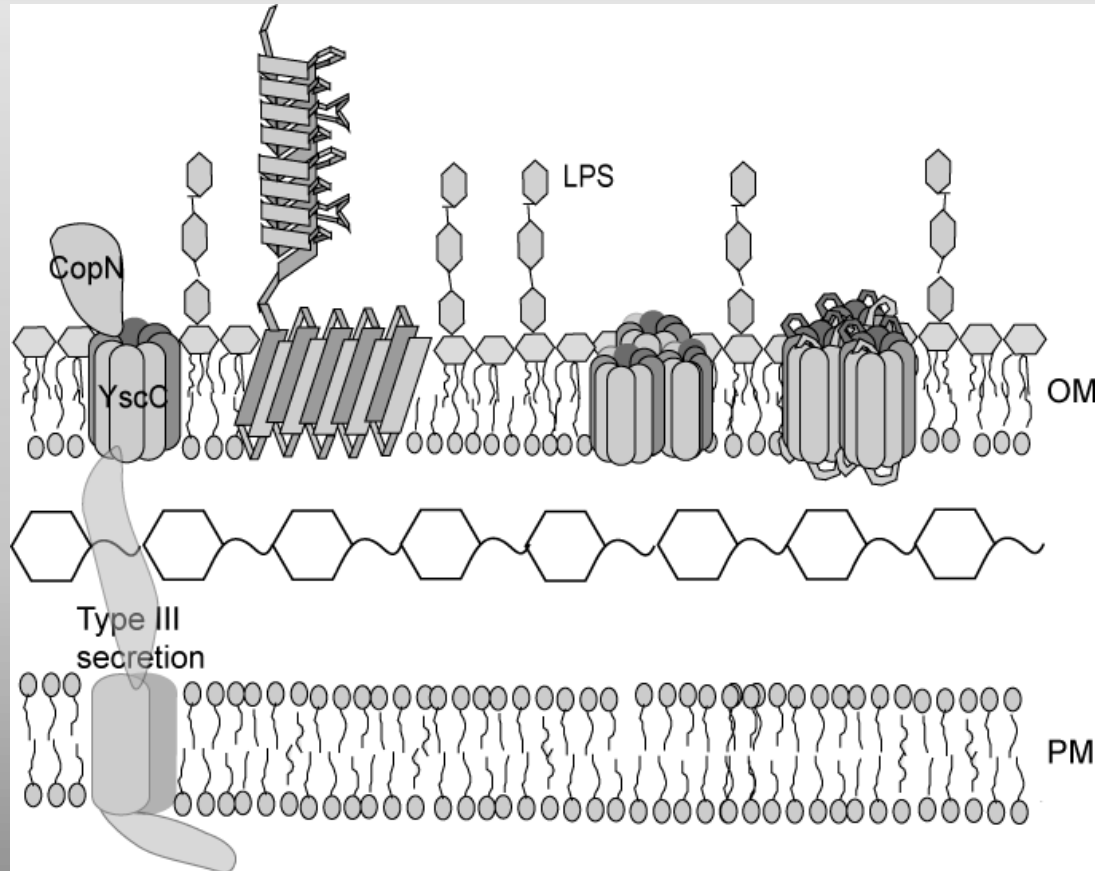


Lipid A

- Lipid A er bevaret i LPS og genkendes af LPS-binding protein og derefter Toll-like receptor 4.
- Dette sætter gang i en kraftig immunreaktion.
- Dvs. LPS er et Pathogen-associated molecular pattern (PAMP) pga. lipid A



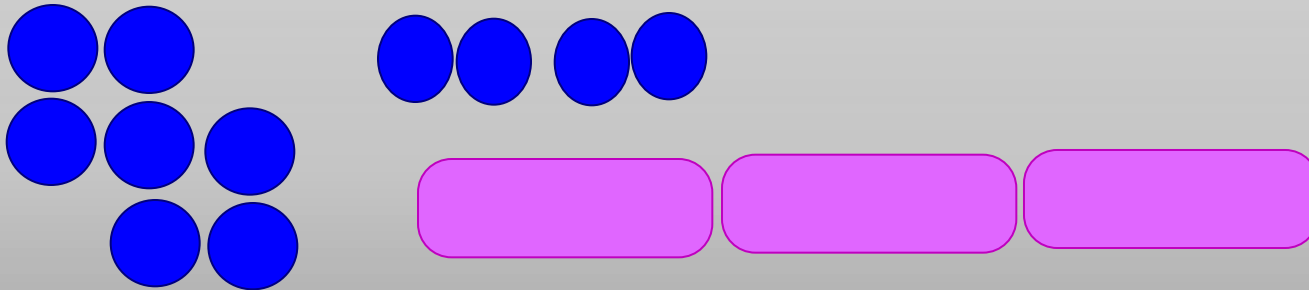
Gram negativ bakterie



Form og lejring af bakterier, [Video](#)

Form og lejring af bakterier, [Video](#)

- Kokker, runde, lancetformet, bønneformet
- Stave, krumme, pleomorfe, små, store
- Klaser, eks. *Staphylococcus aureus*
- Kæder, eks. *Streptococcus pneumoniae*



Lejring kæder

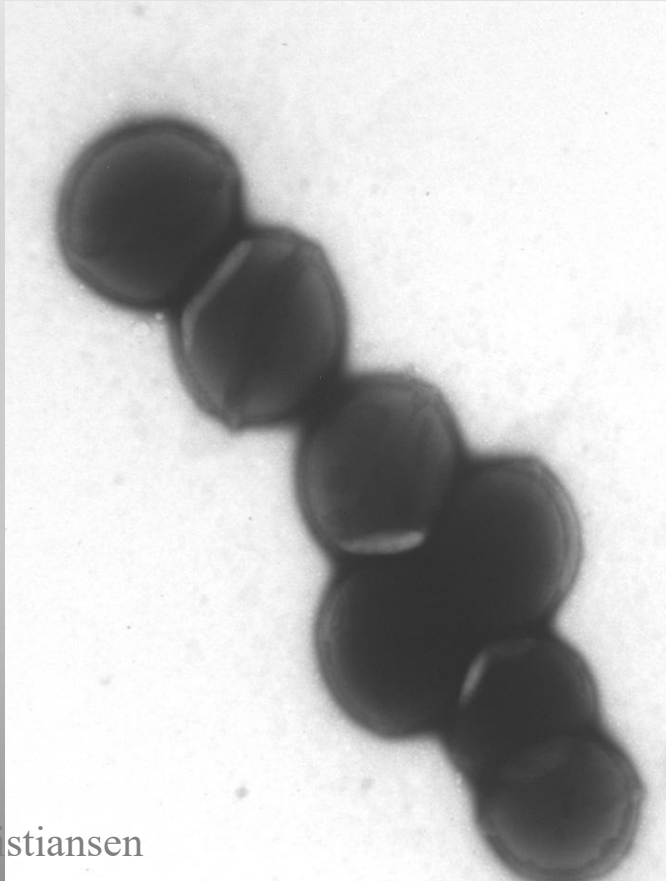
Streptococcus pneumoniae



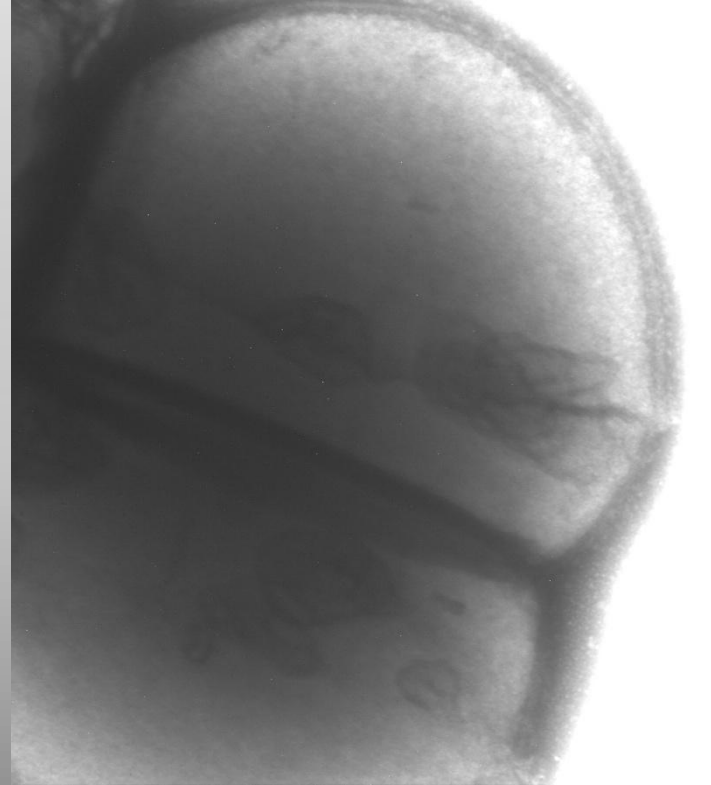
Streptococcus viridans



Streptococcus viridans, EM

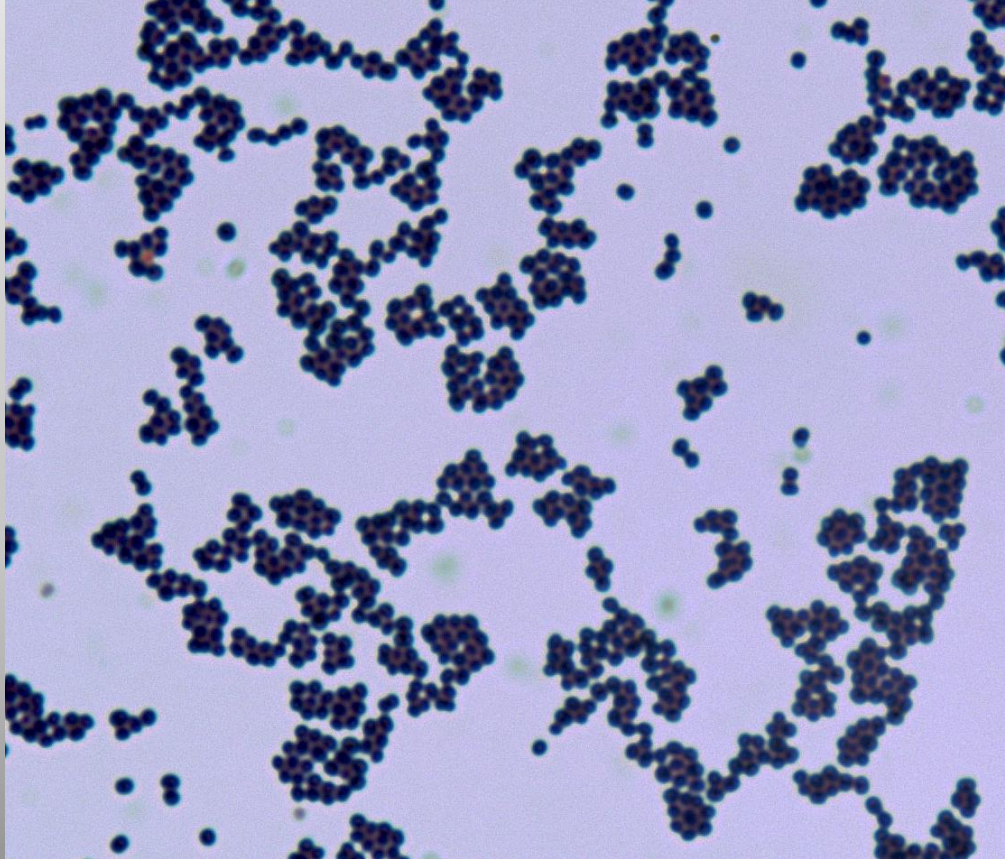


Negativ farvning



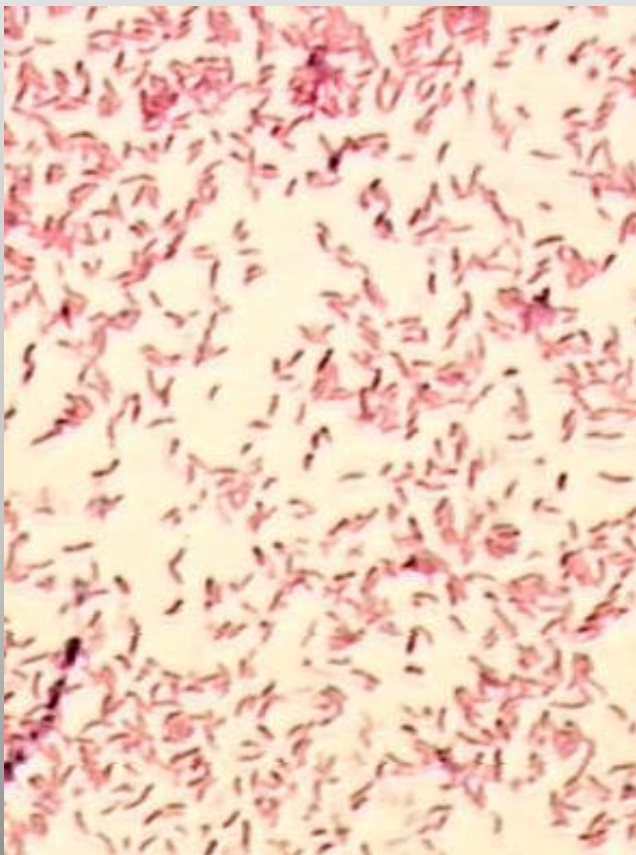
Lejring klaser

Staphylococcus aureus

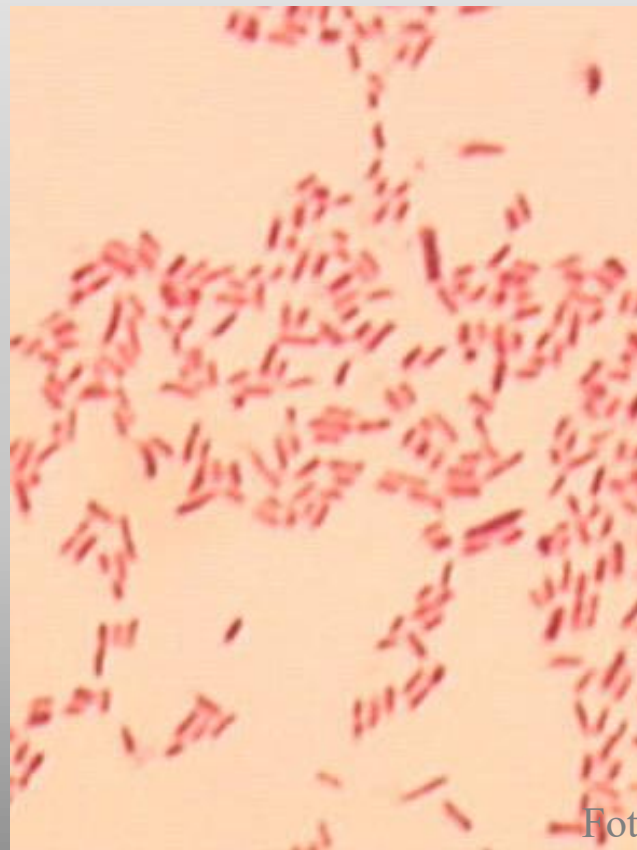


Stave: form størrelse

Vibrio cholerae

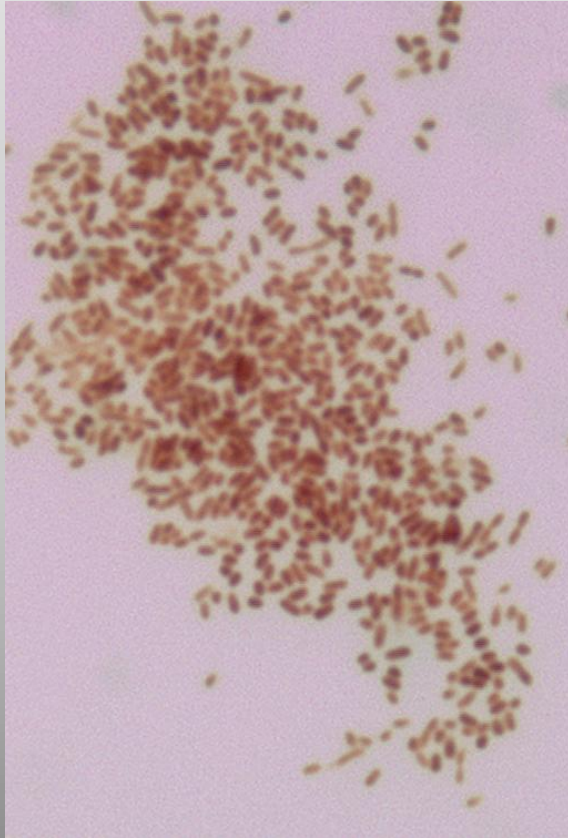


Escherichia coli



Stave: form størrelse

Haemophilus influenzae



Bacillus cereus



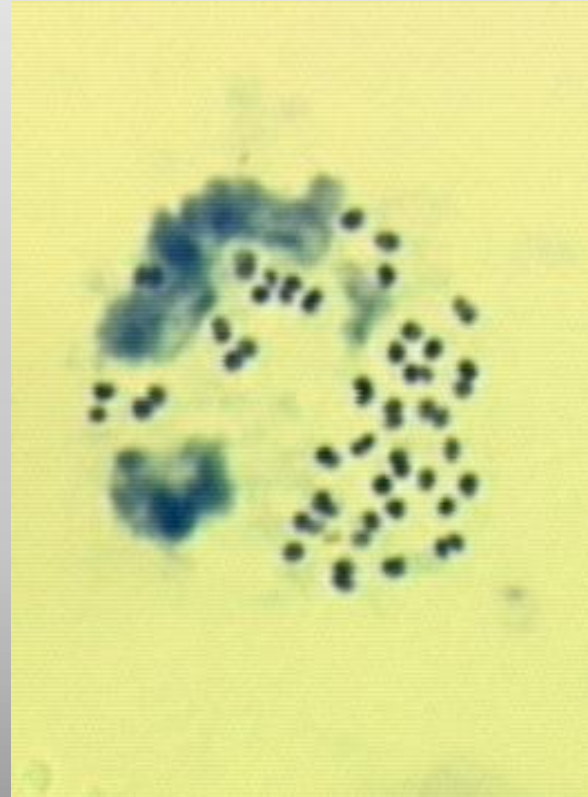
Form af bakterier

Diplokokker (side by side)



Foto: S.Birkelund

Gramfarvning



Metylenblåtfarving

Diplokokker (end to end)

Gram farvning, lysmikroskop



Foto: S.Birkelund

Negativ farvning , EM

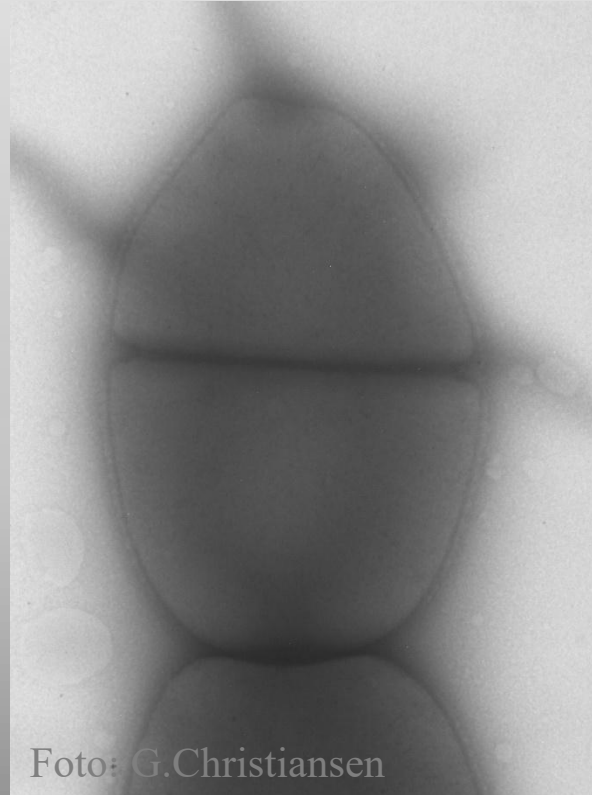


Foto: G.Christiansen

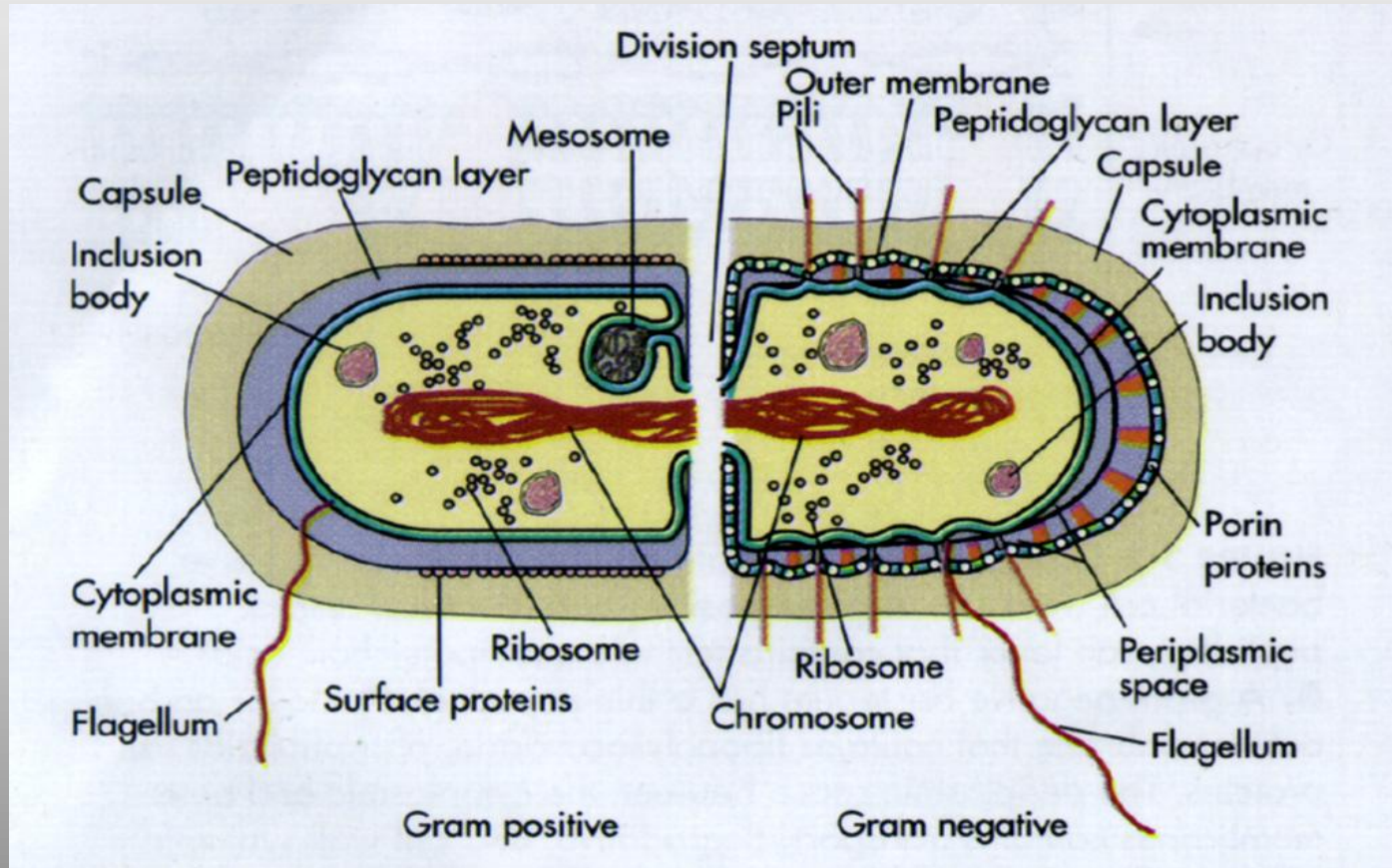
Streptococcus pneumoniae

Organelles

Organelles

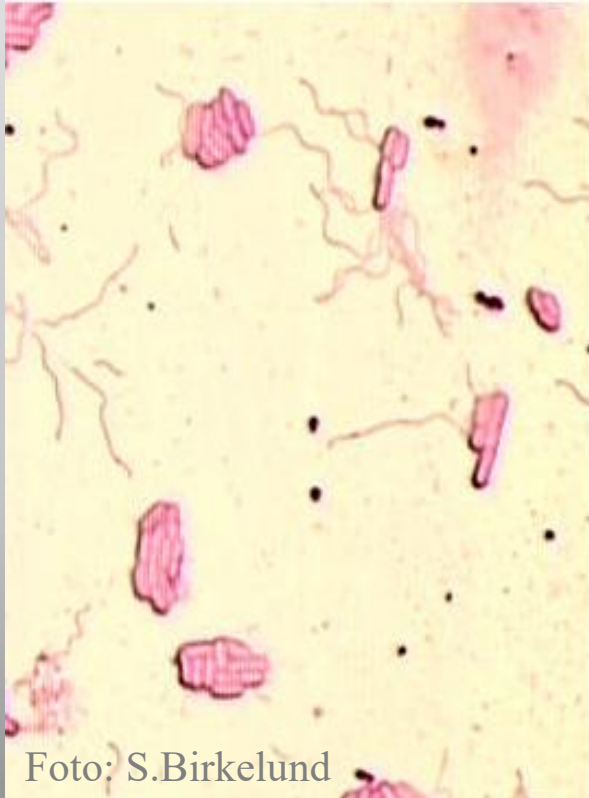
Gram positiv

Gram negativ



Flageller, *Proteus mirabilis*

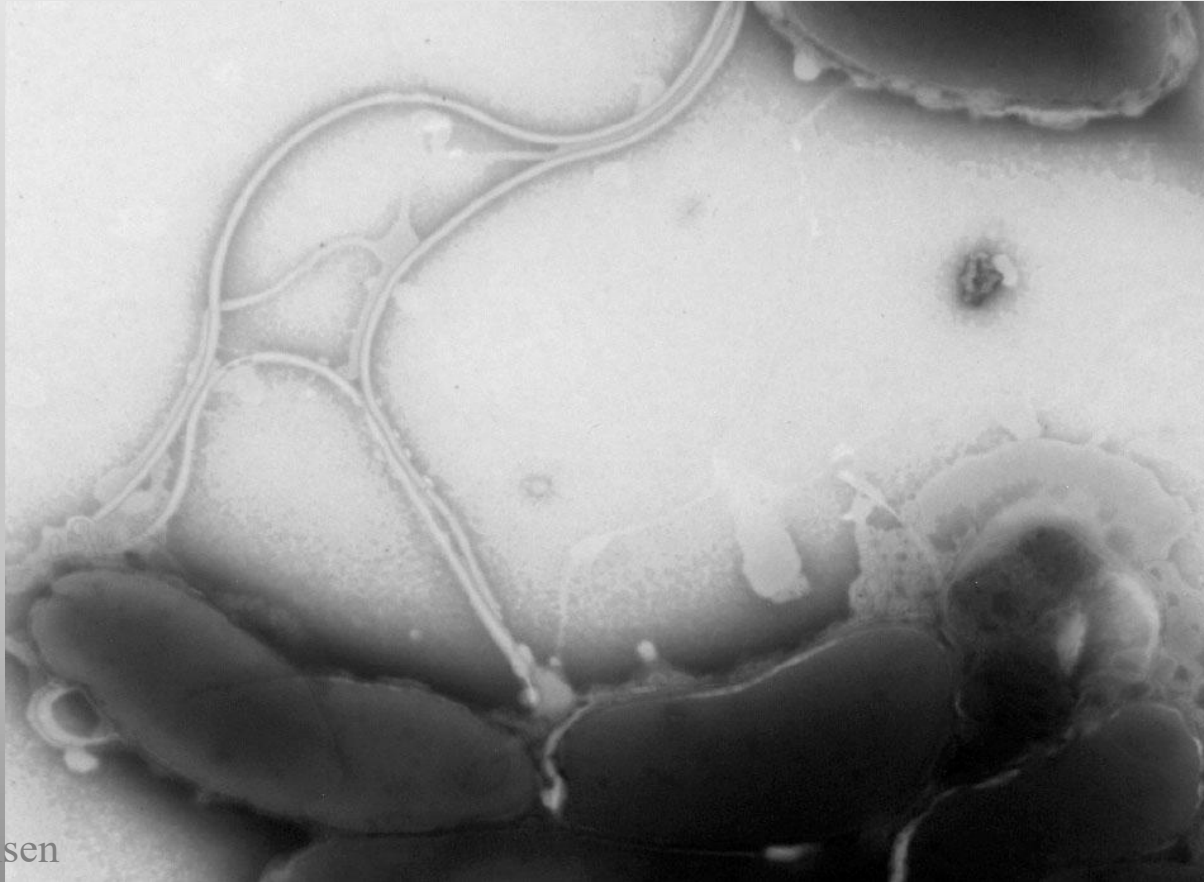
Flagelfarvning



EM negativfarvning

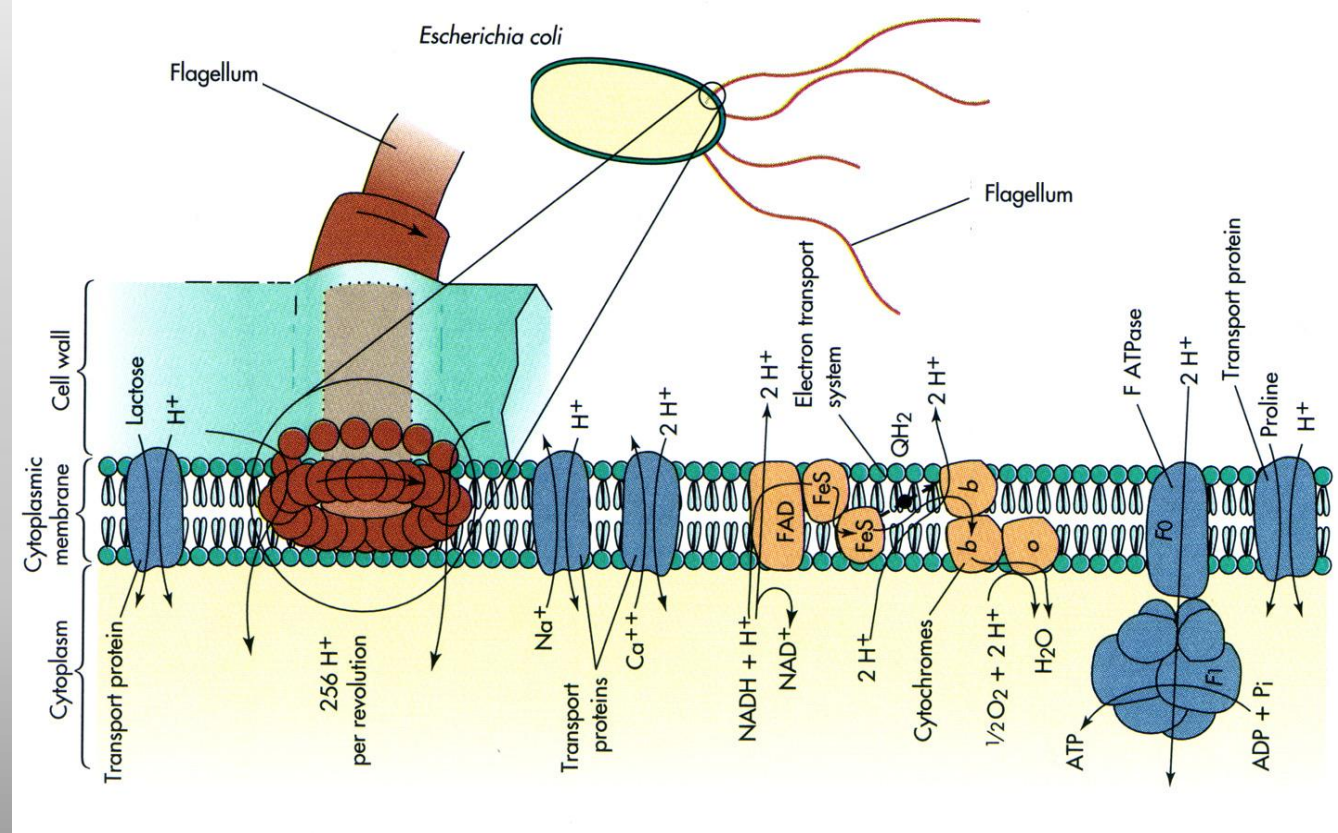


Flageller *Vibrio cholerae*



Model for flagelmotilitet

Af human
pathogene
bakterier er
det kun gram
negative der
har flageller



Pili / Fimbriae

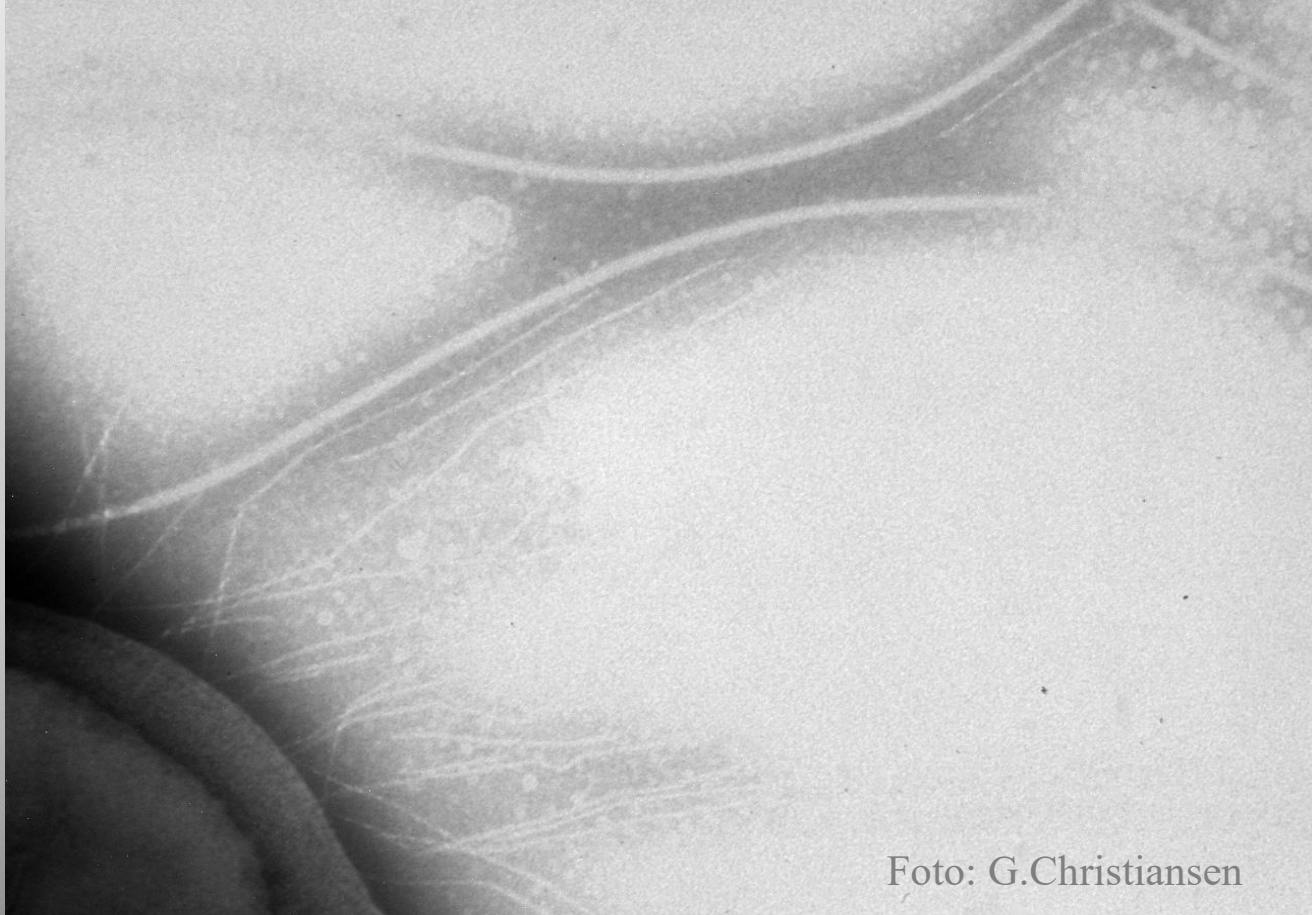
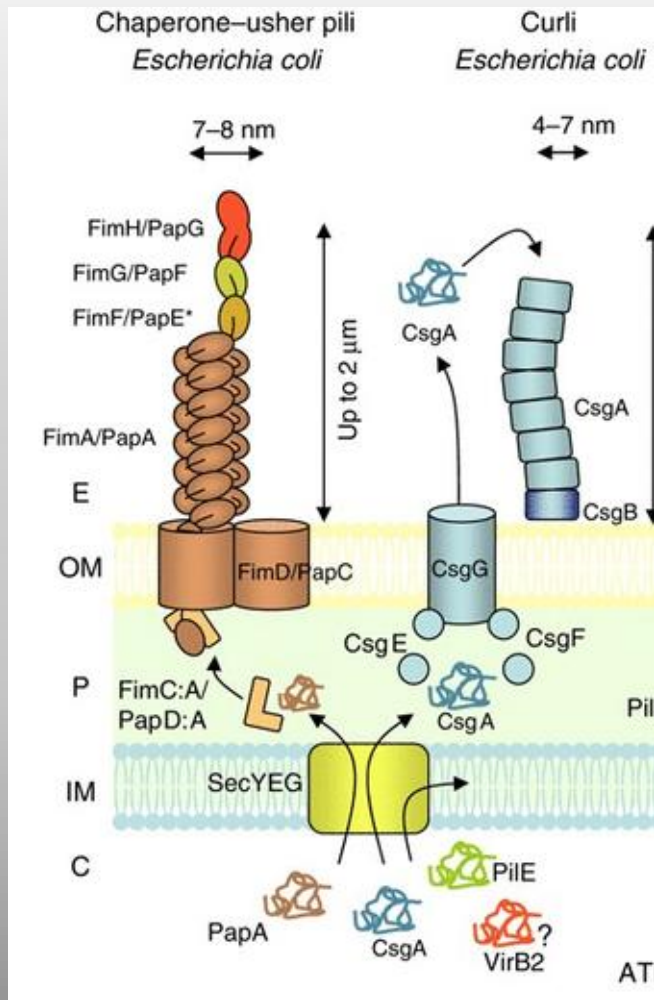


Foto: G.Christiansen



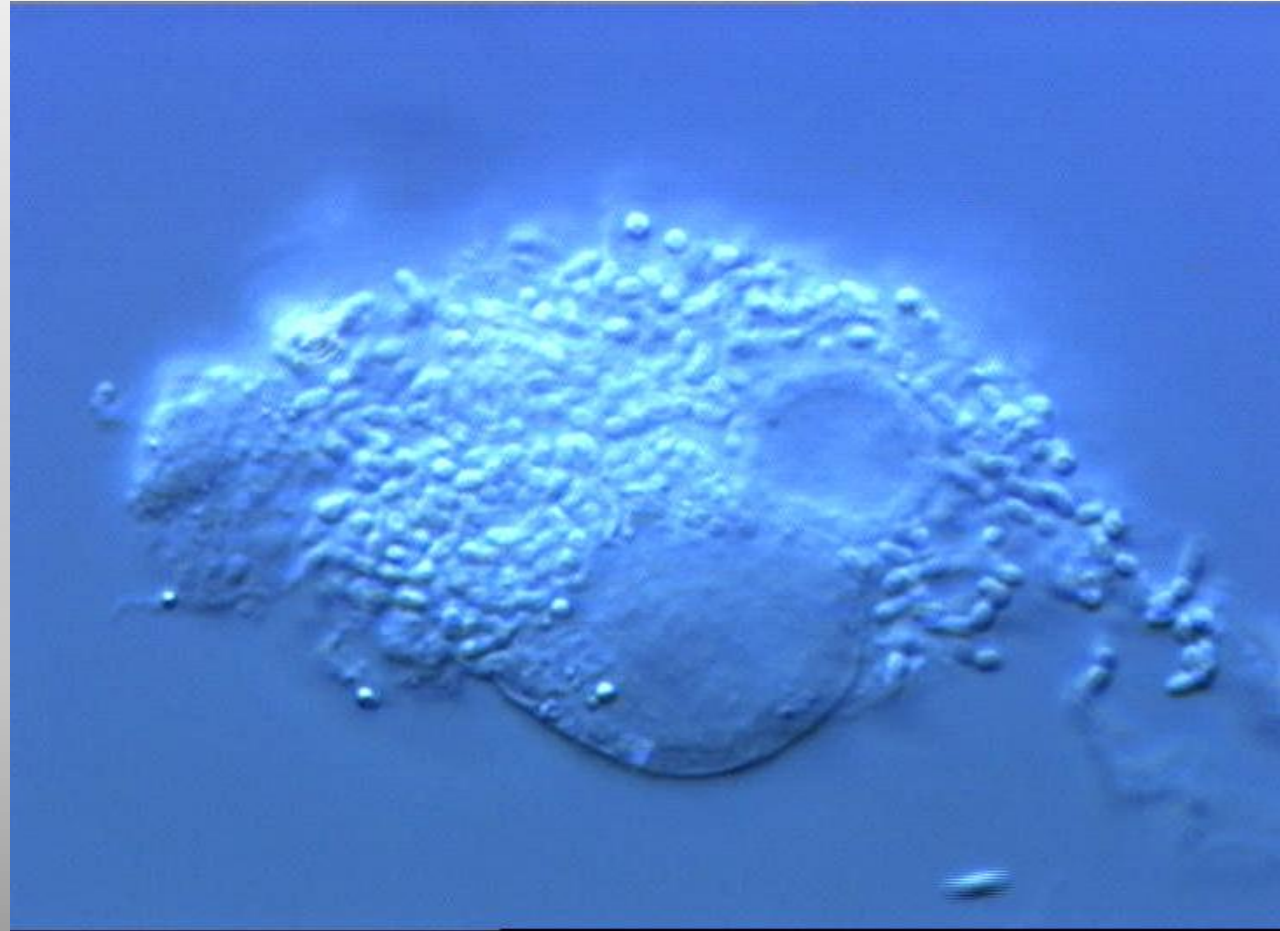
Pili er ikke motile Er kun hæftet i ydermembranen

Foto: G.Christiansen

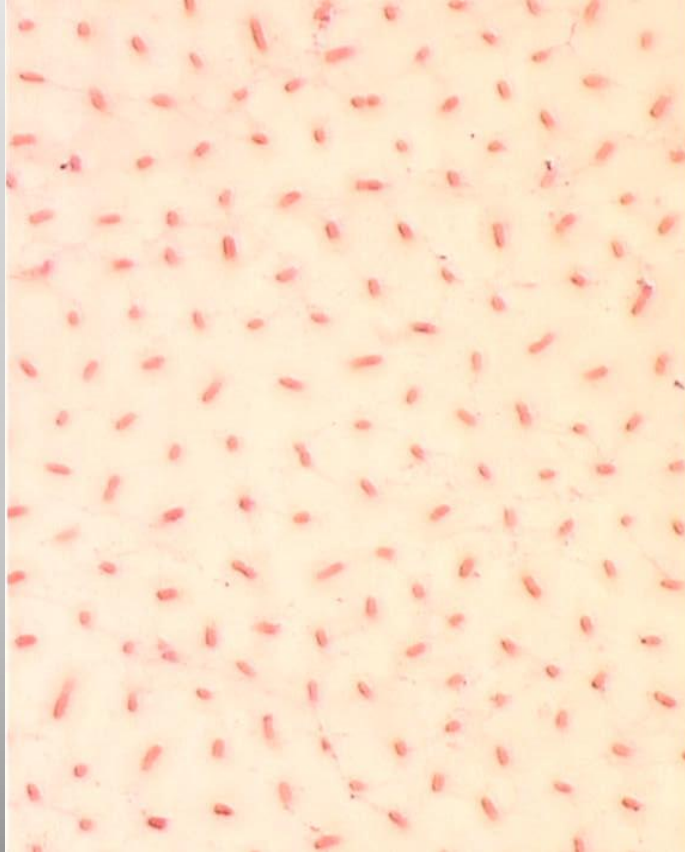


- Pili bruges til
adhærence til celler
og overflader

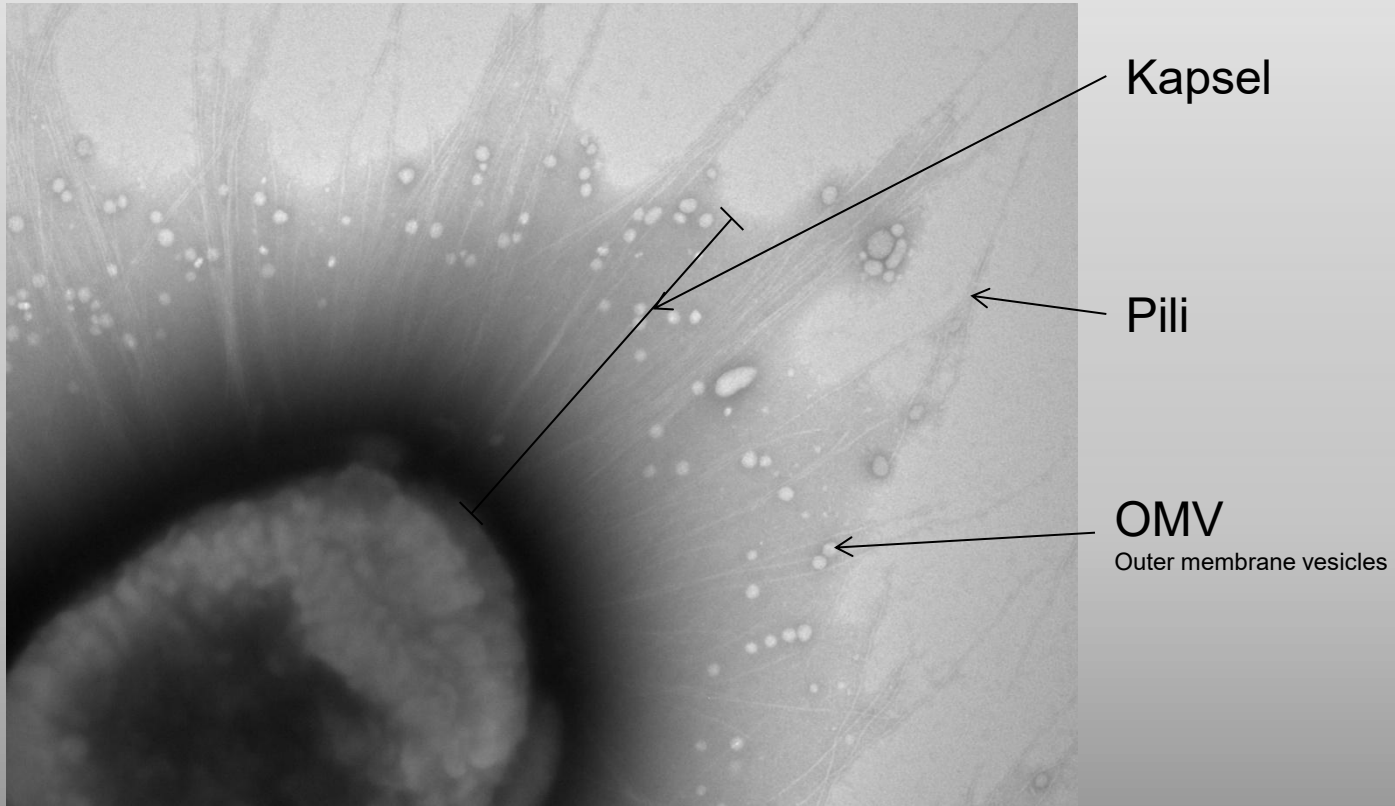
Differential
interferens kontrast
(DIC) mikroskopi af
urin fra patient med
E. coli cystitis.



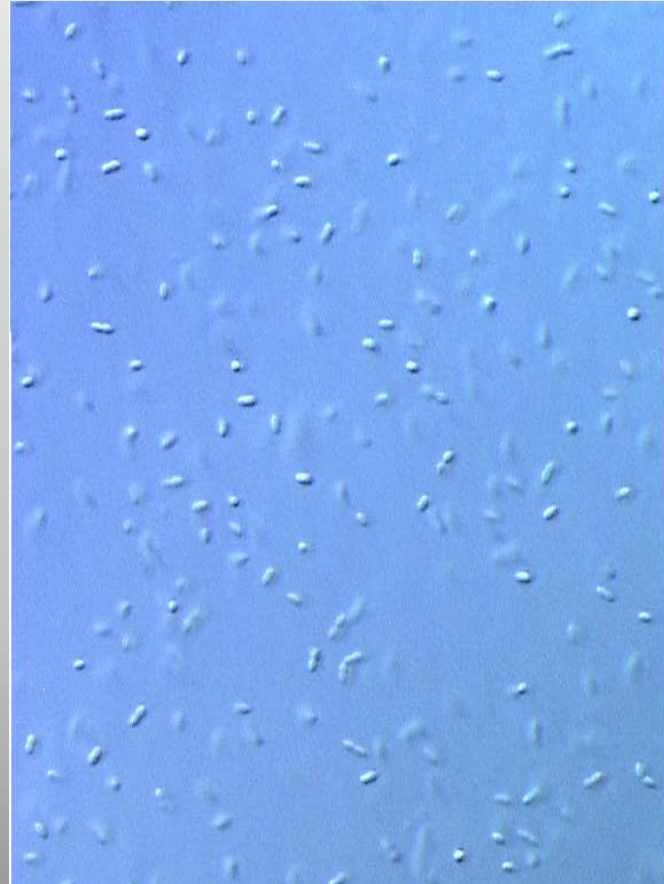
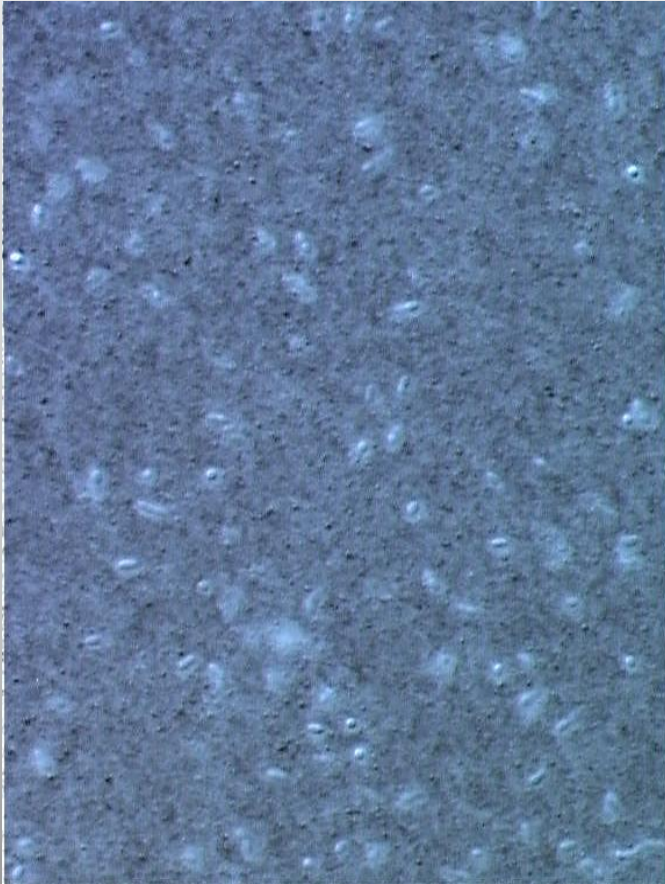
Kapsel, *Klebsiella pneumoniae*



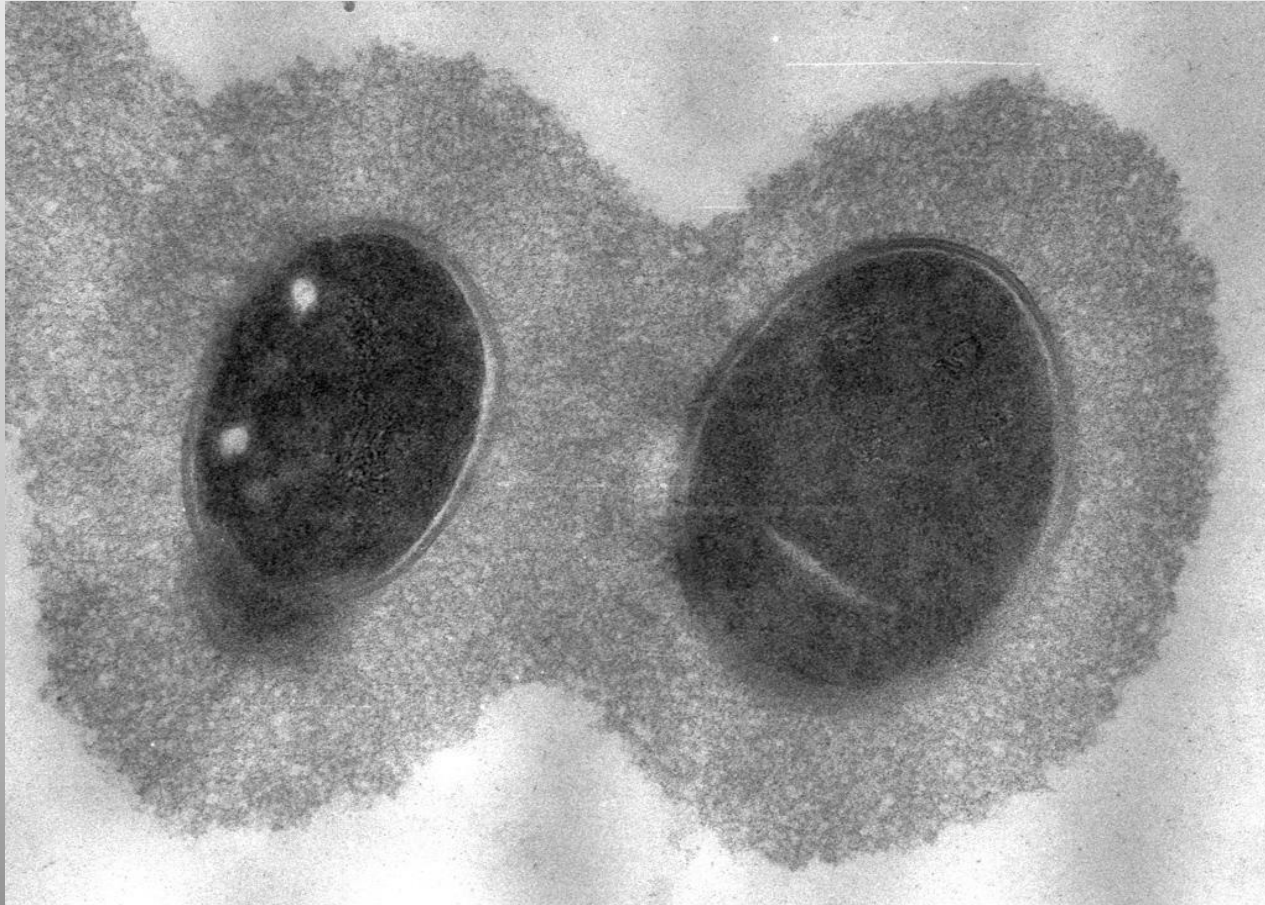
Klebsiella pneumoniae



Streptococcus pneumoniae



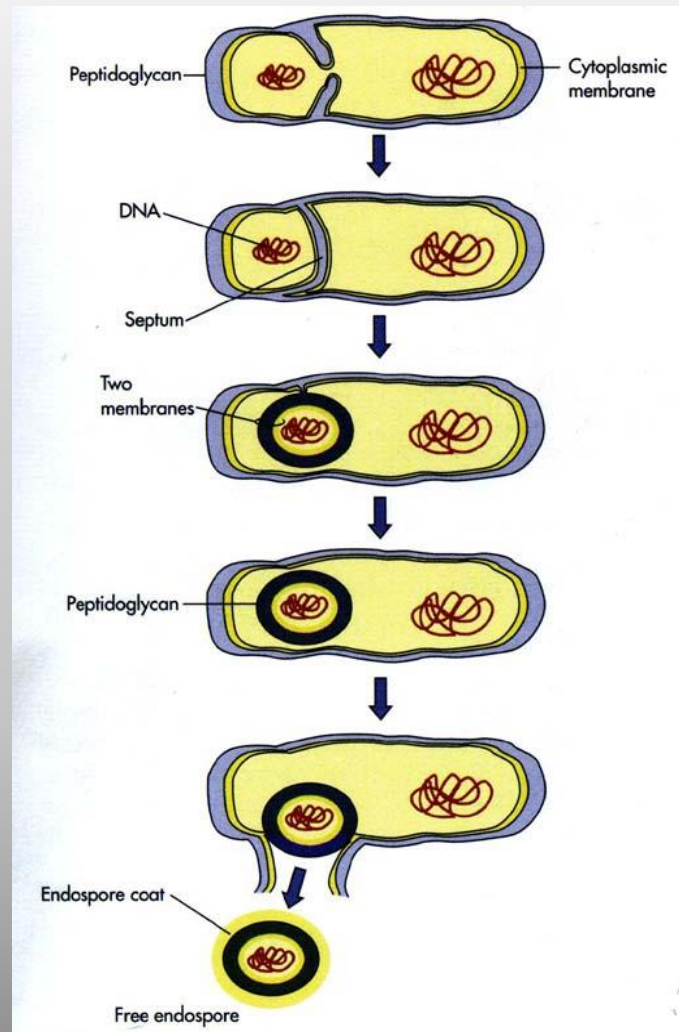
S. pneumoniae (EM, Uffe Skov)



Sporer

Sporer
er fysiks
meget stabile

inaktiveres
ved
autoklaving

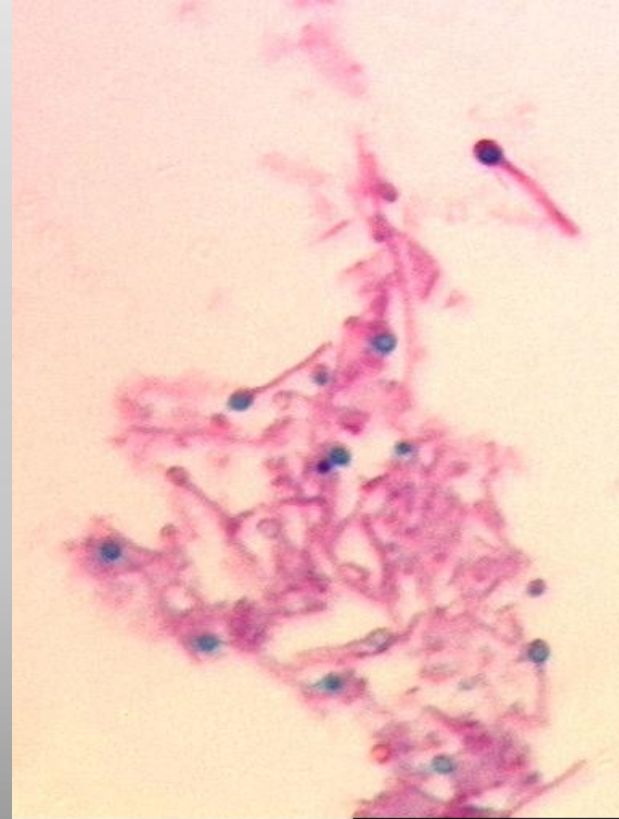


Sporer

Bacillus anthracis



Clostridium tetani



Slut